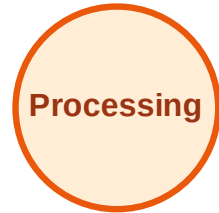


BFS Traversal

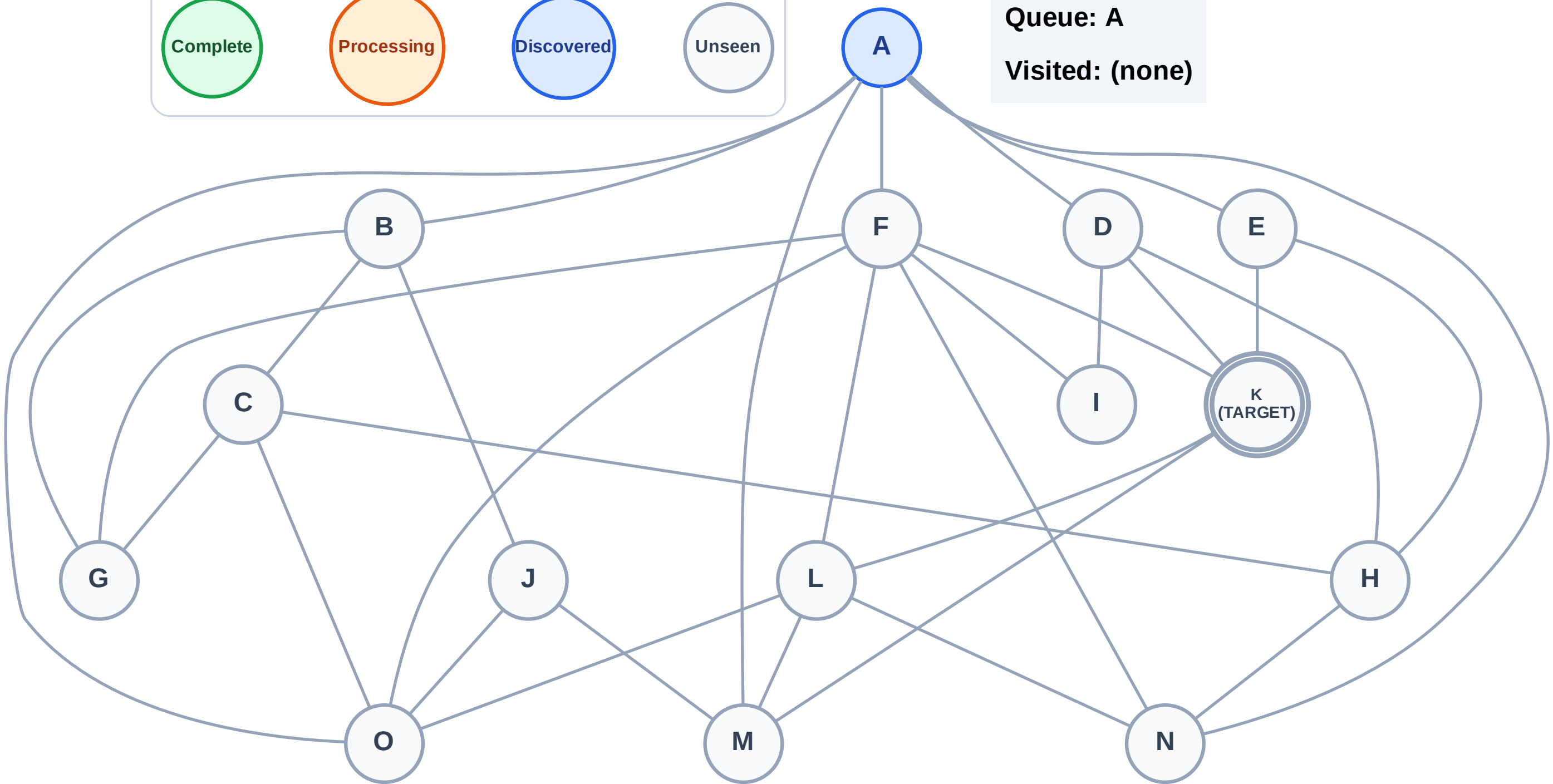
Step 1: Start at A (Seeking target node K)

Legend



Queue: A

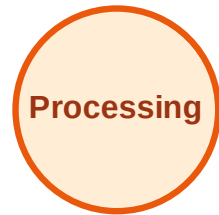
Visited: (none)



BFS Traversal

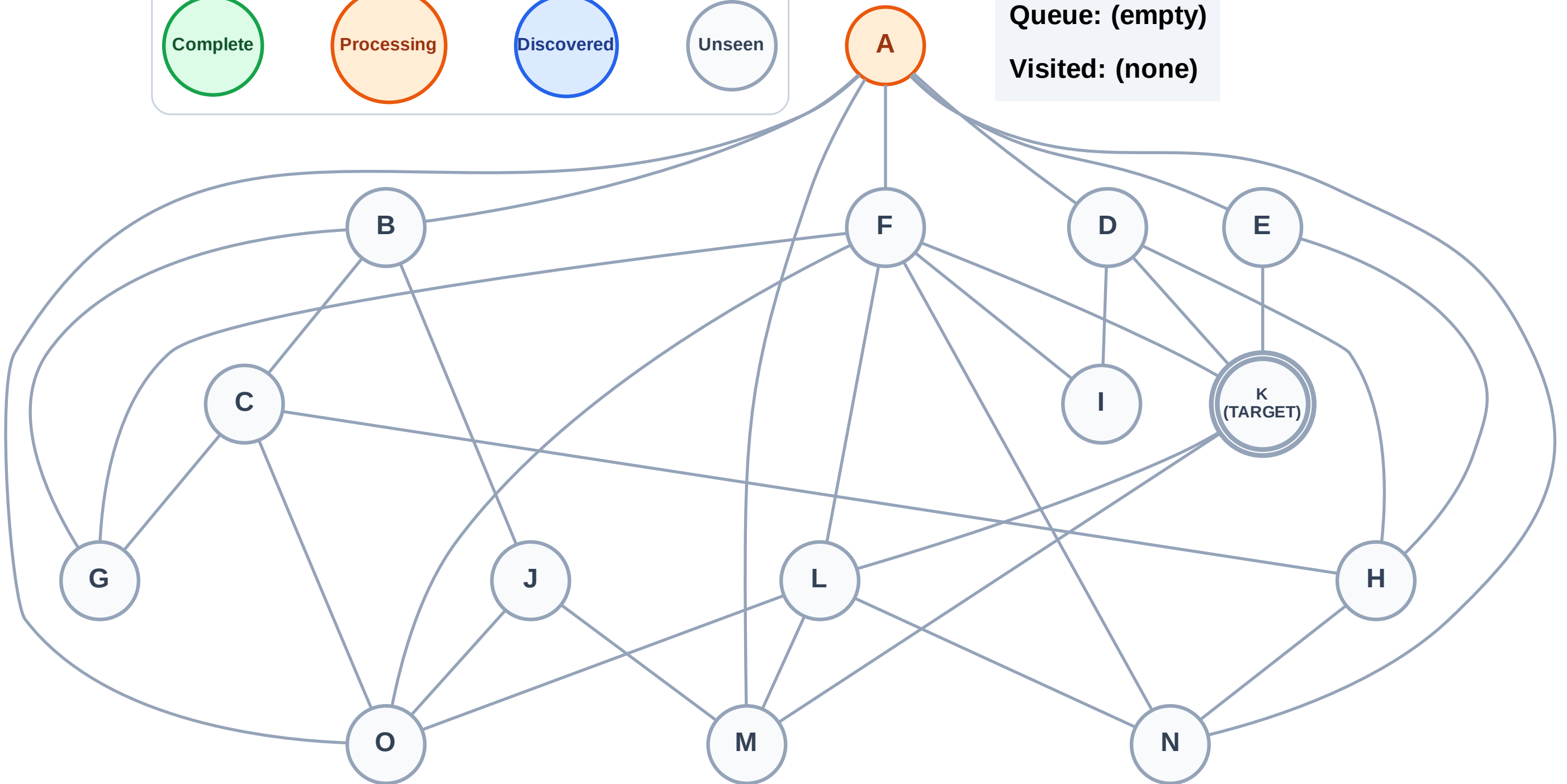
Step 2: Process node A

Legend



Queue: (empty)

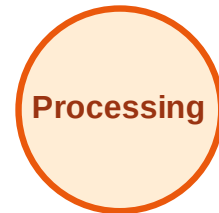
Visited: (none)



BFS Traversal

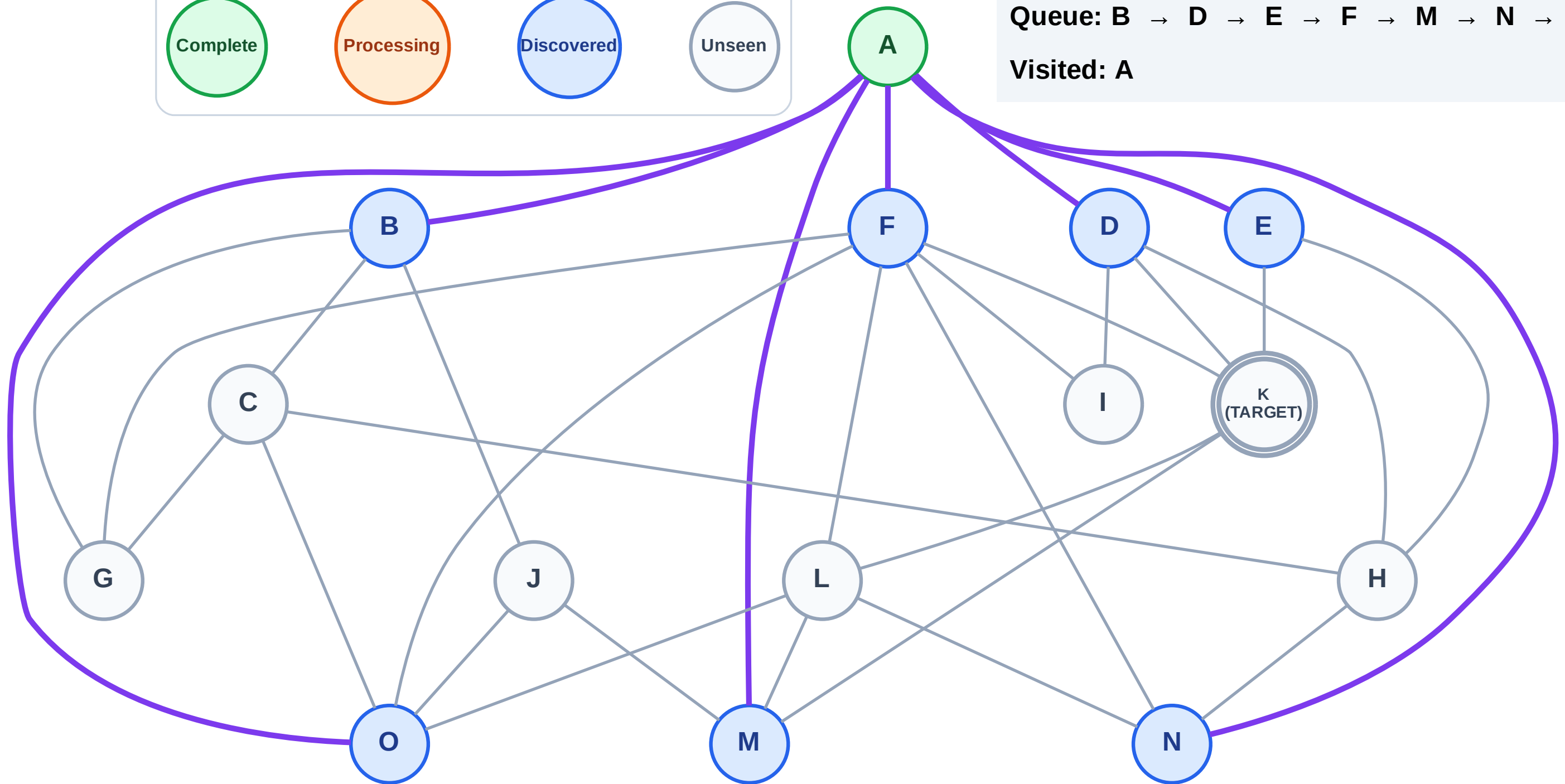
Step 3: Discover B, D, E, F, M, N, O from A

Legend



Queue: B → D → E → F → M → N → O

Visited: A



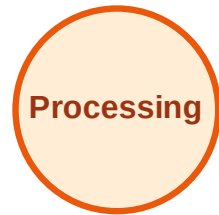
BFS Traversal

Step 4: Process node B

Legend



Complete



Processing



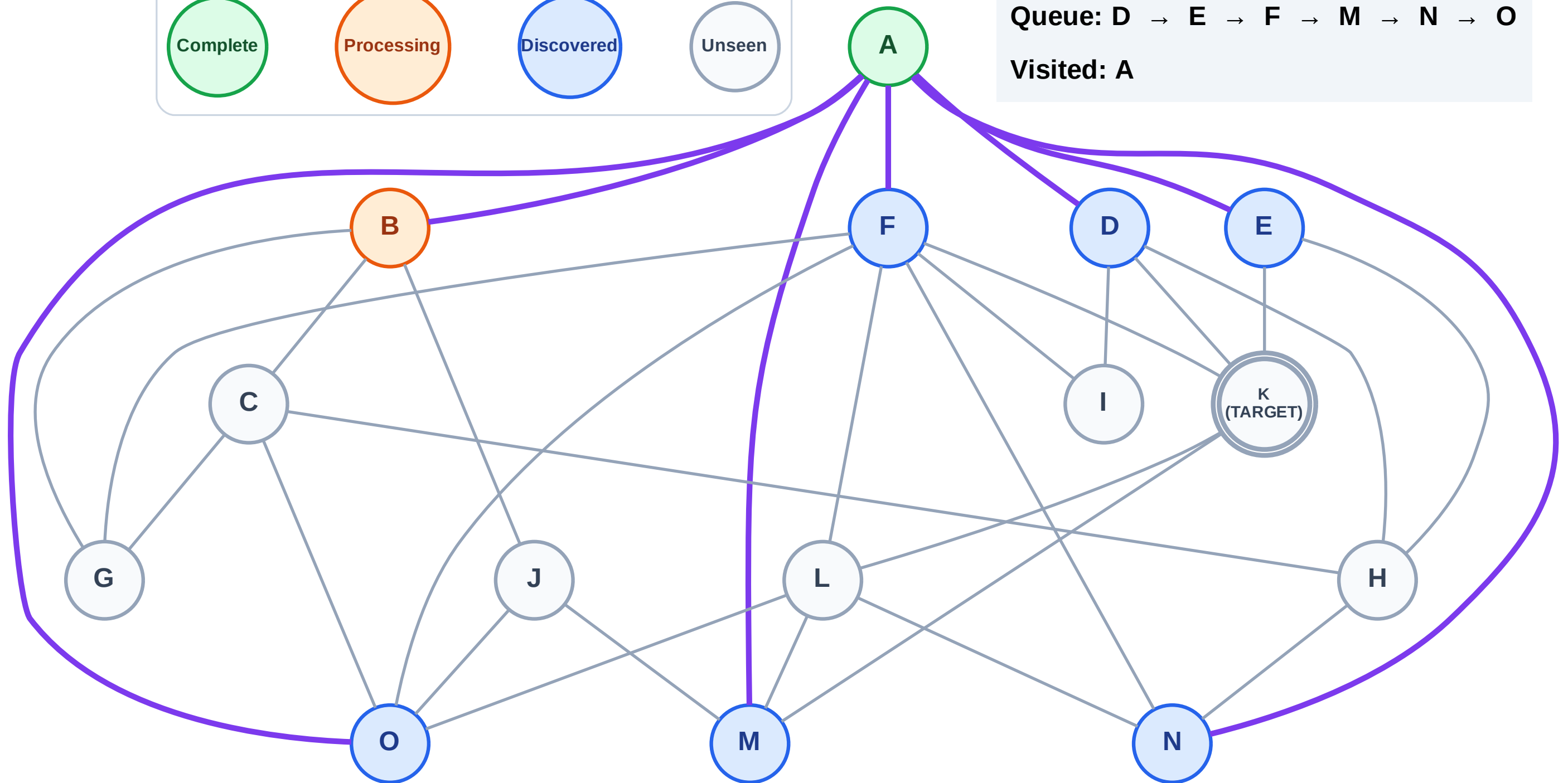
Discovered



Unseen

Queue: D → E → F → M → N → O

Visited: A



BFS Traversal

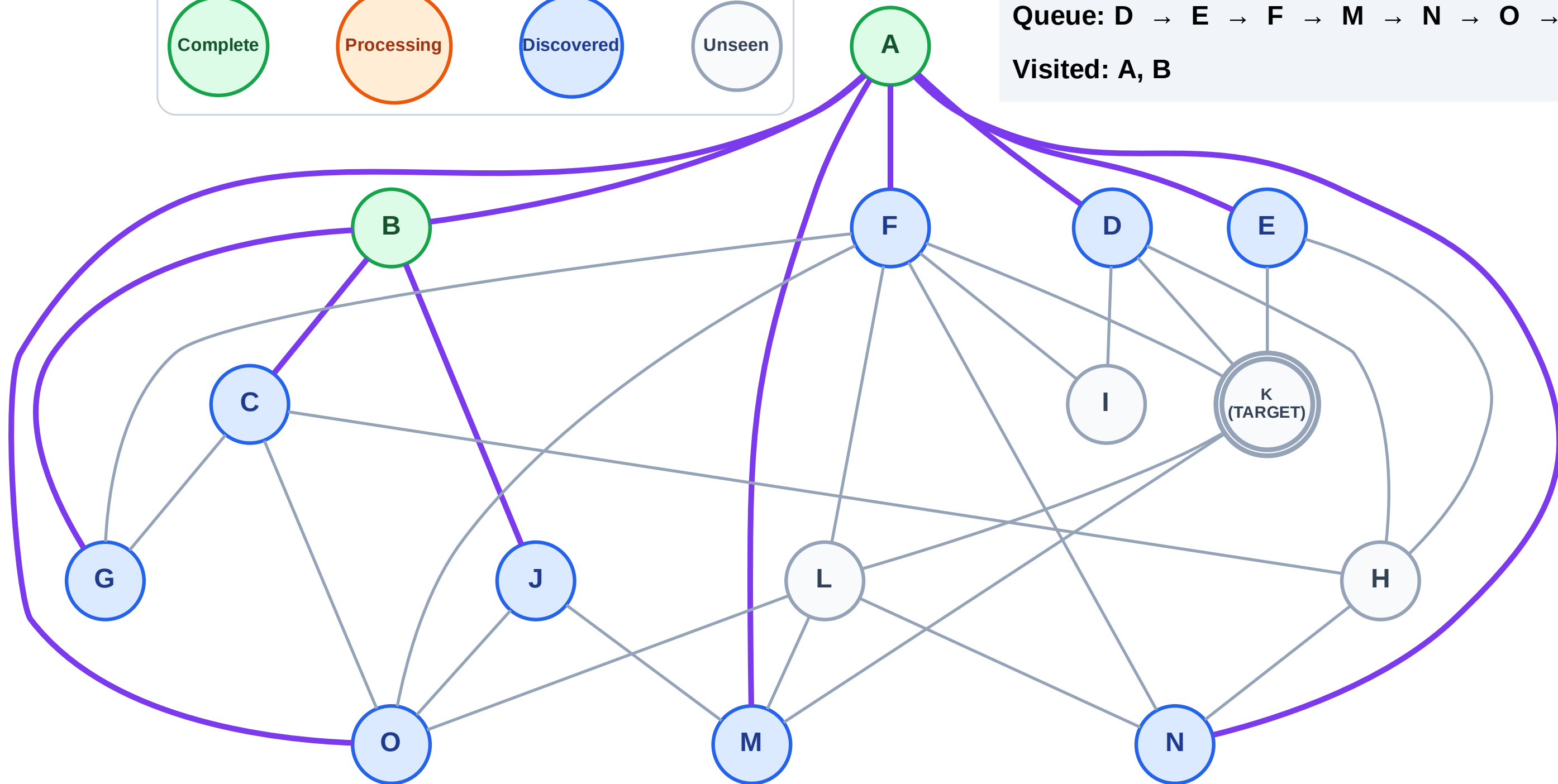
Step 5: Discover C, G, J from B

Legend



Queue: D → E → F → M → N → O → C → G → J

Visited: A, B



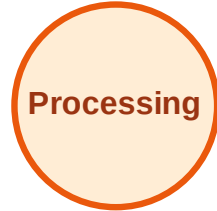
BFS Traversal

Step 6: Process node D

Legend



Complete



Processing



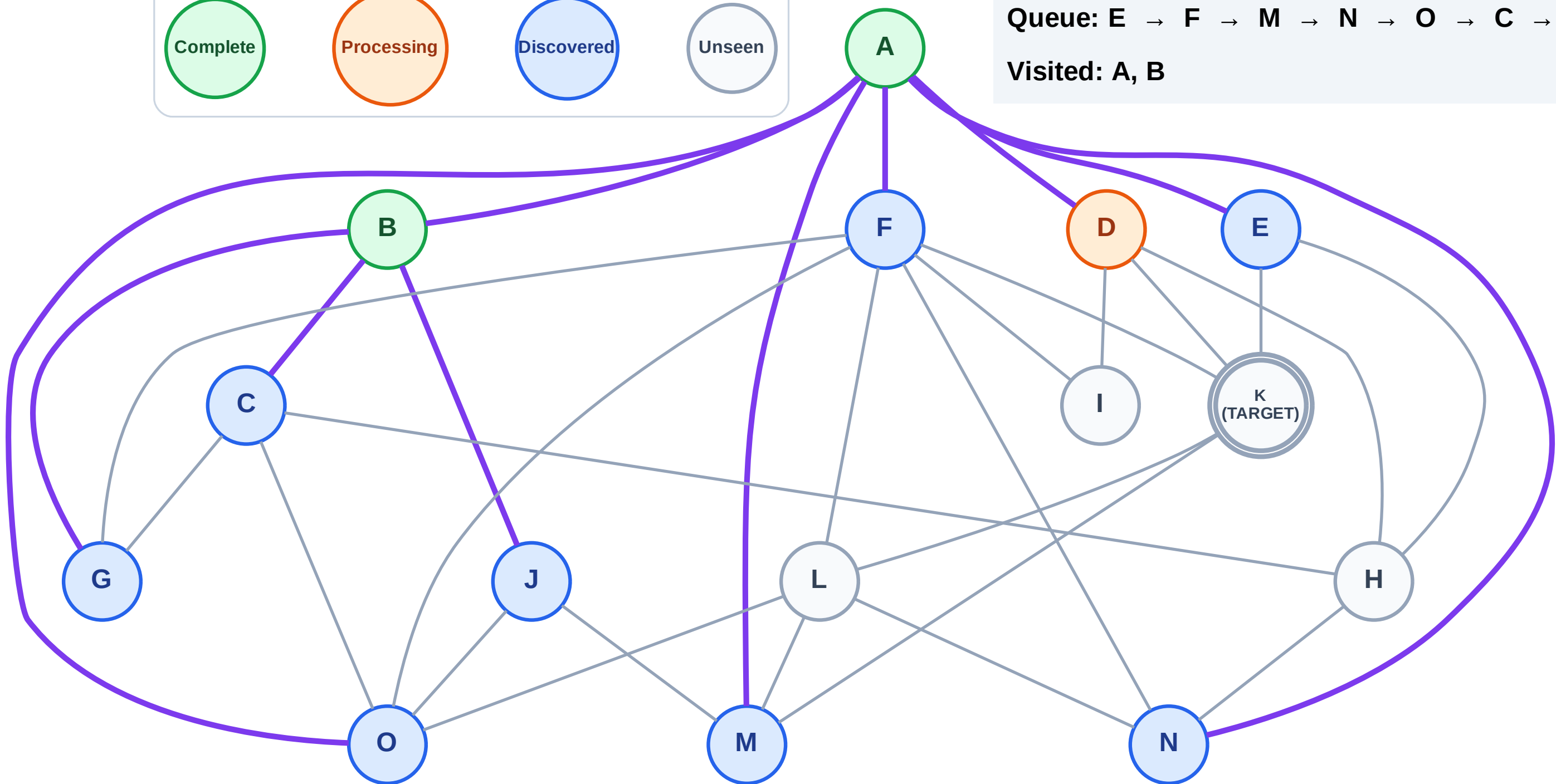
Discovered



Unseen

Queue: E → F → M → N → O → C → G → J

Visited: A, B



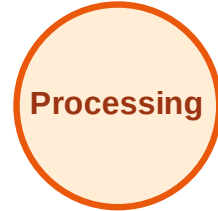
BFS Traversal

Step 7: Discover H, I, K from D

Legend



Complete



Processing



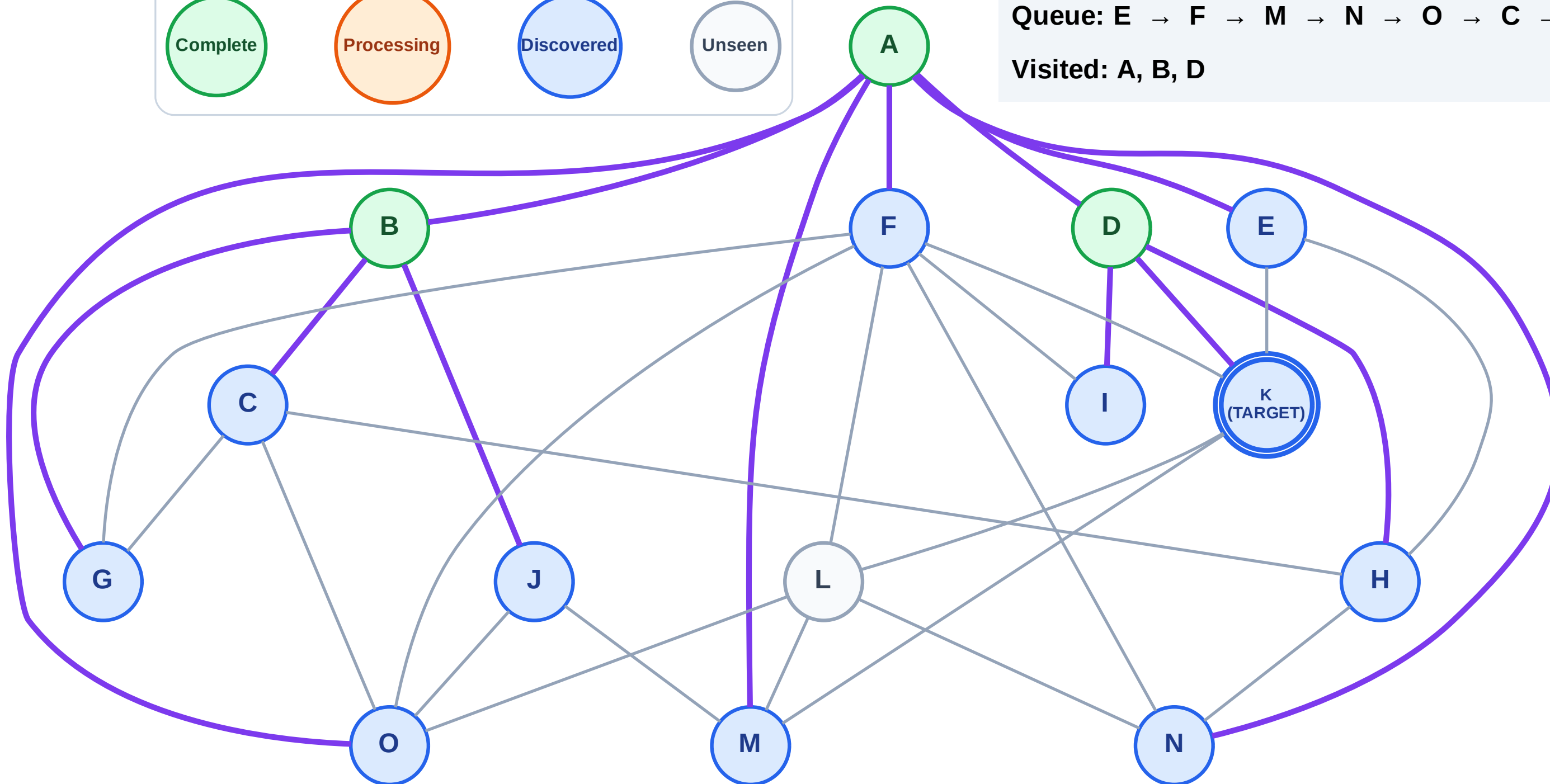
Discovered



Unseen

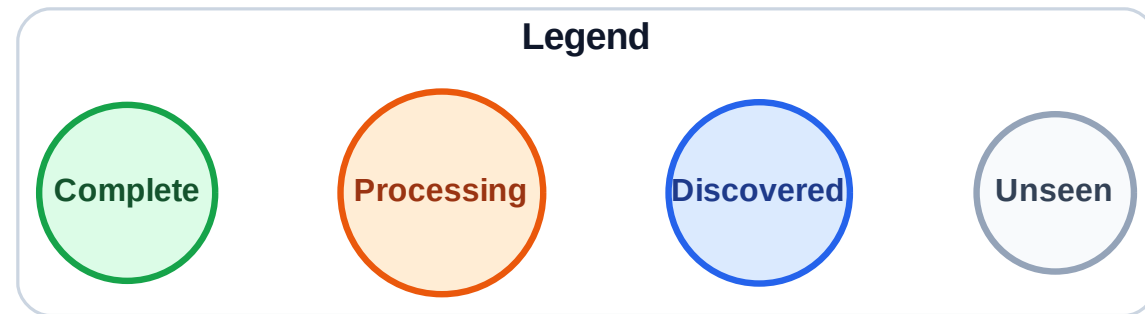
Queue: E → F → M → N → O → C → G → J → H → I → K

Visited: A, B, D



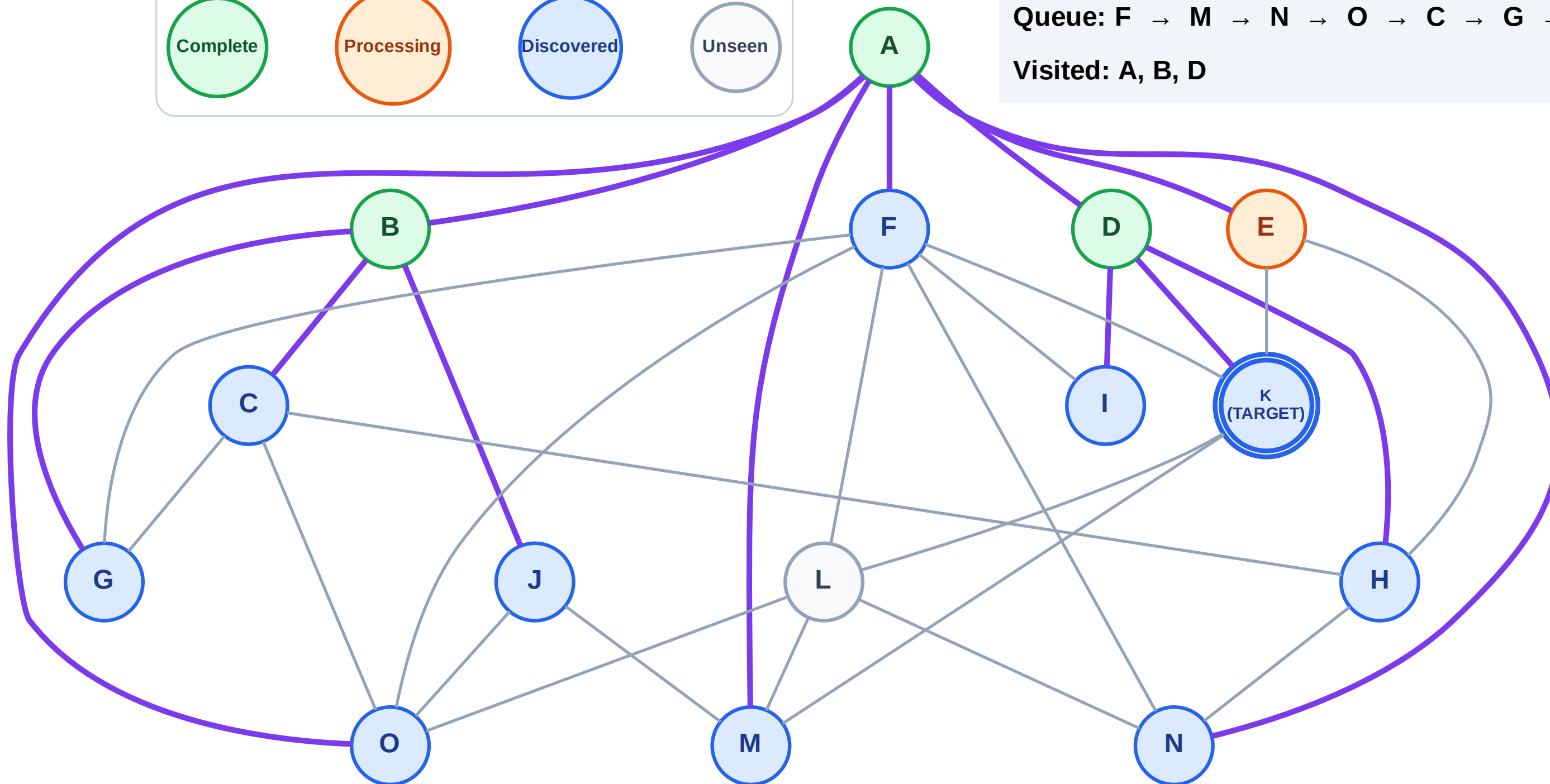
BFS Traversal

Step 8: Process node E



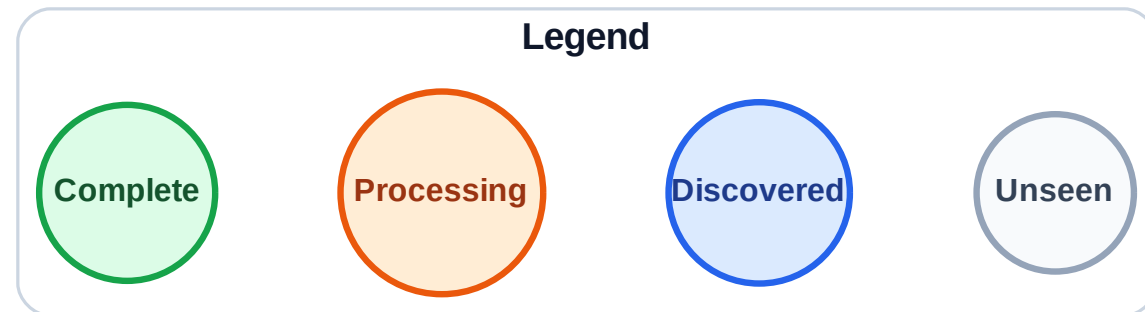
Queue: F → M → N → O → C → G → J → H → I → K

Visited: A, B, D



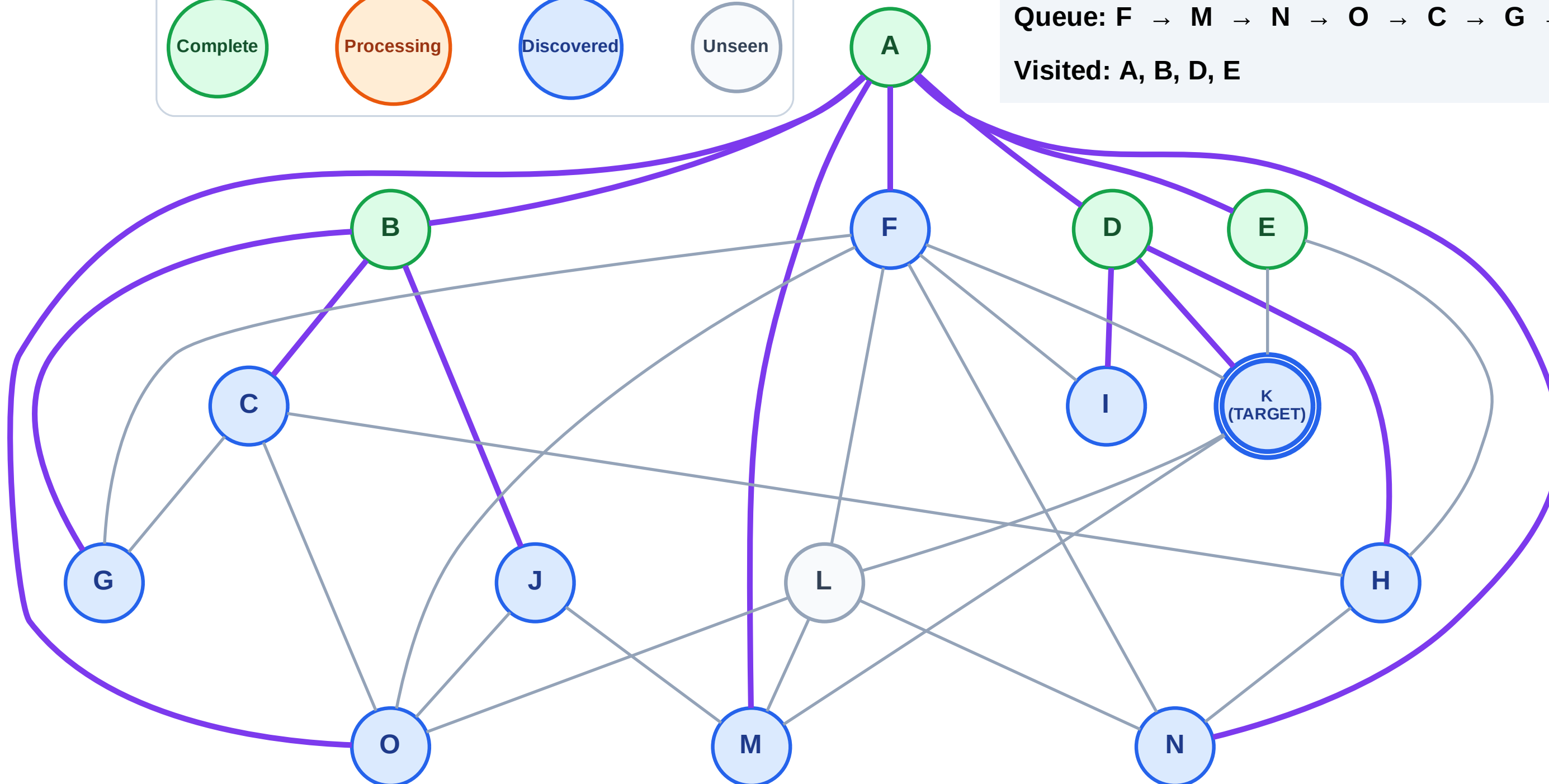
BFS Traversal

Step 9: No new neighbors from E



Queue: F → M → N → O → C → G → J → H → I → K

Visited: A, B, D, E



BFS Traversal

Step 10: Process node F

Legend

Complete

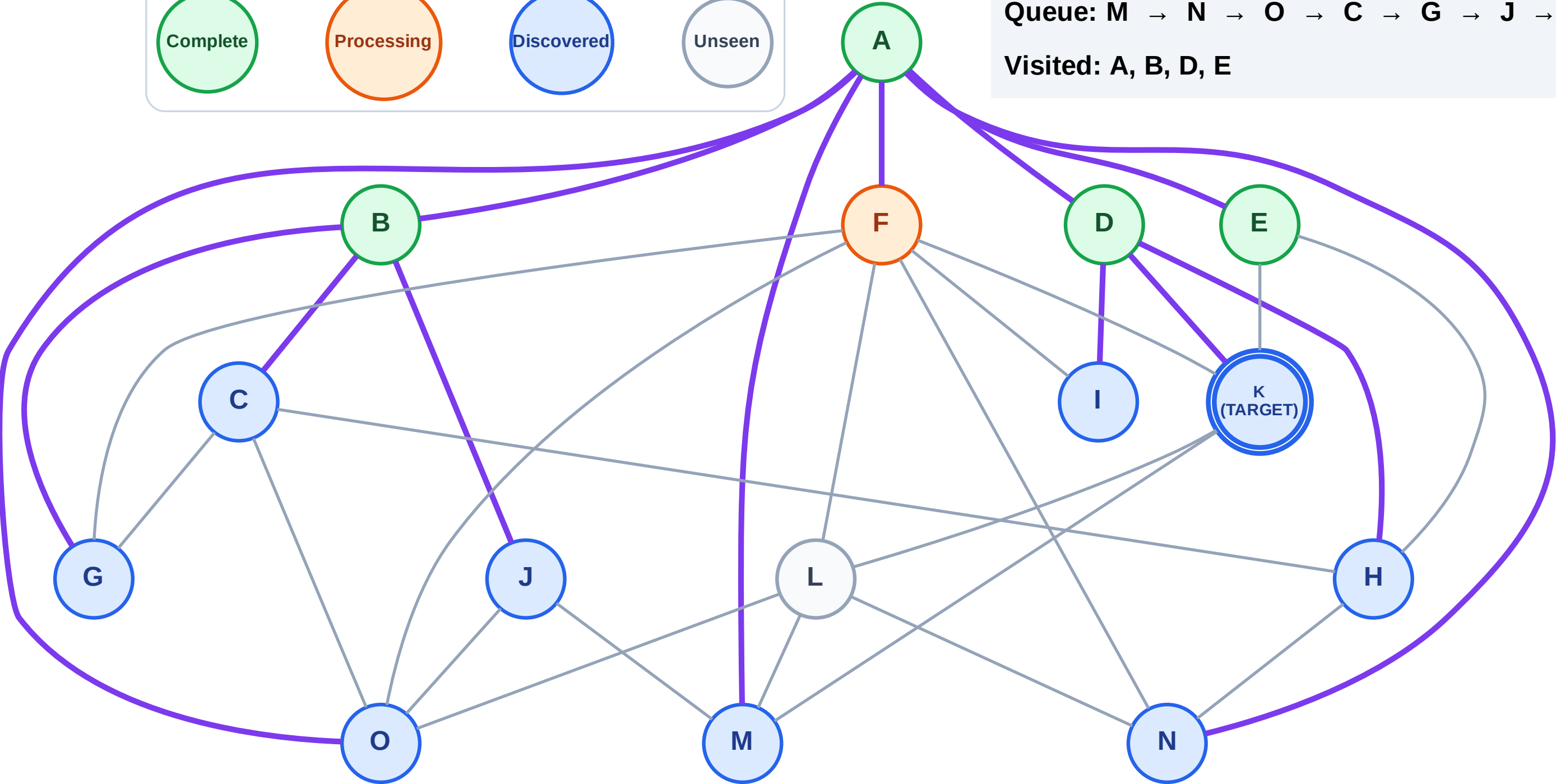
Processing

Discovered

Unseen

Queue: M → N → O → C → G → J → H → I → K

Visited: A, B, D, E



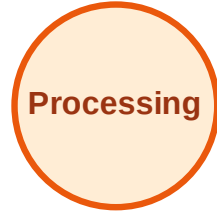
BFS Traversal

Step 11: Discover L from F

Legend



Complete



Processing



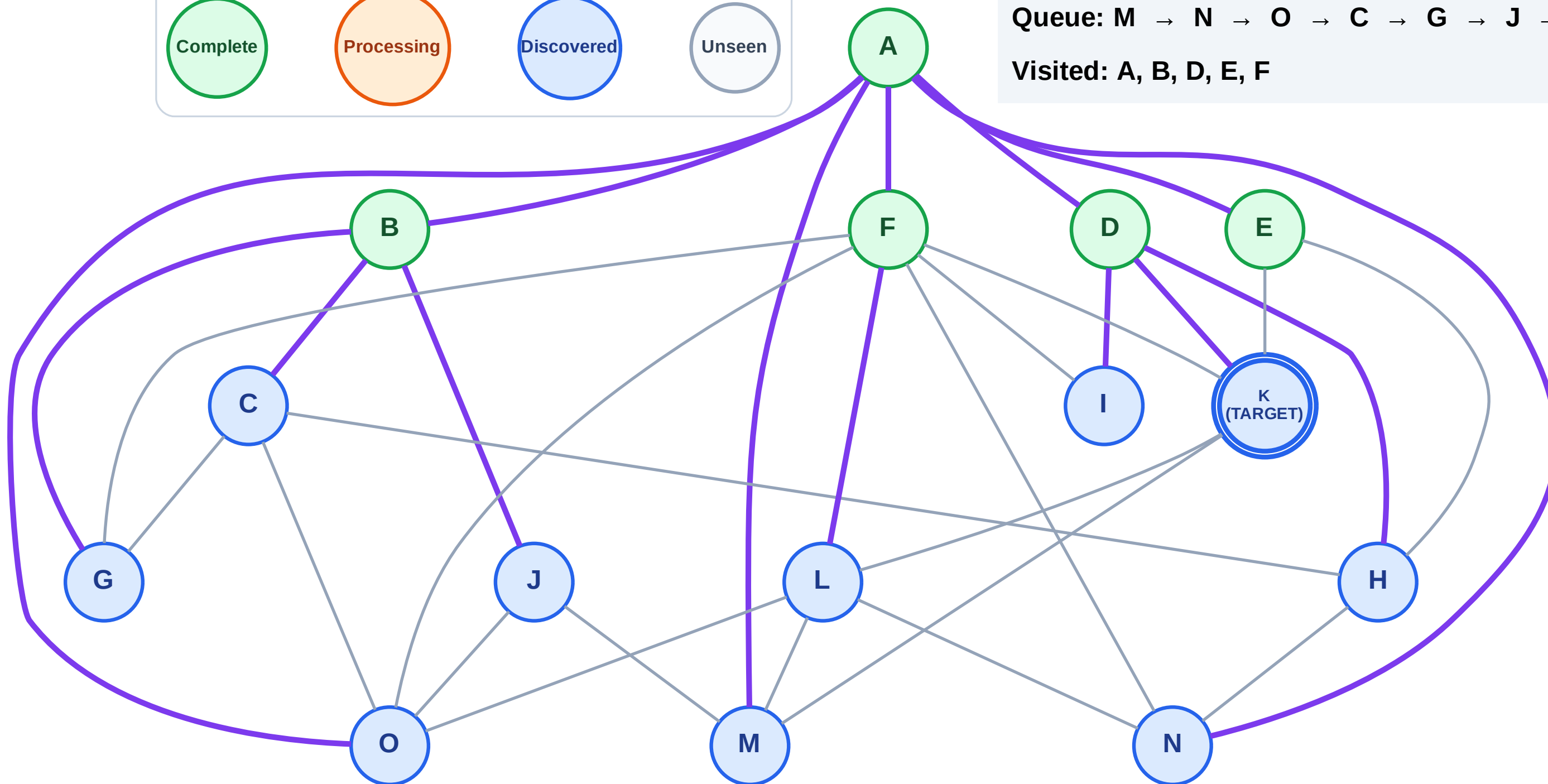
Discovered



Unseen

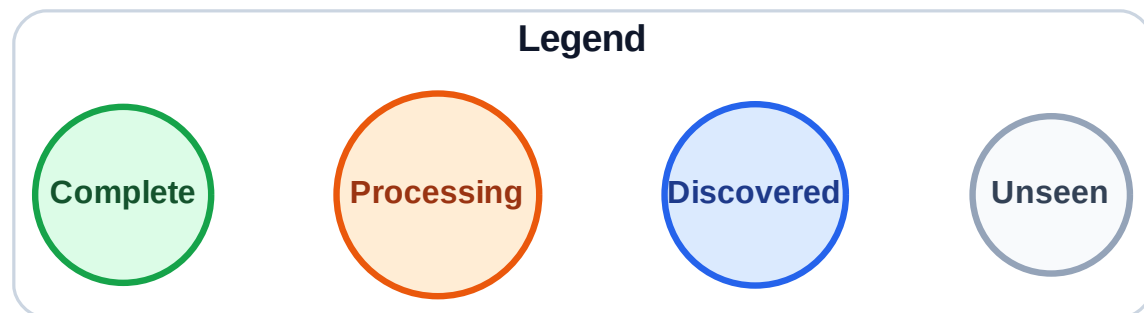
Queue: M → N → O → C → G → J → H → I → K → L

Visited: A, B, D, E, F



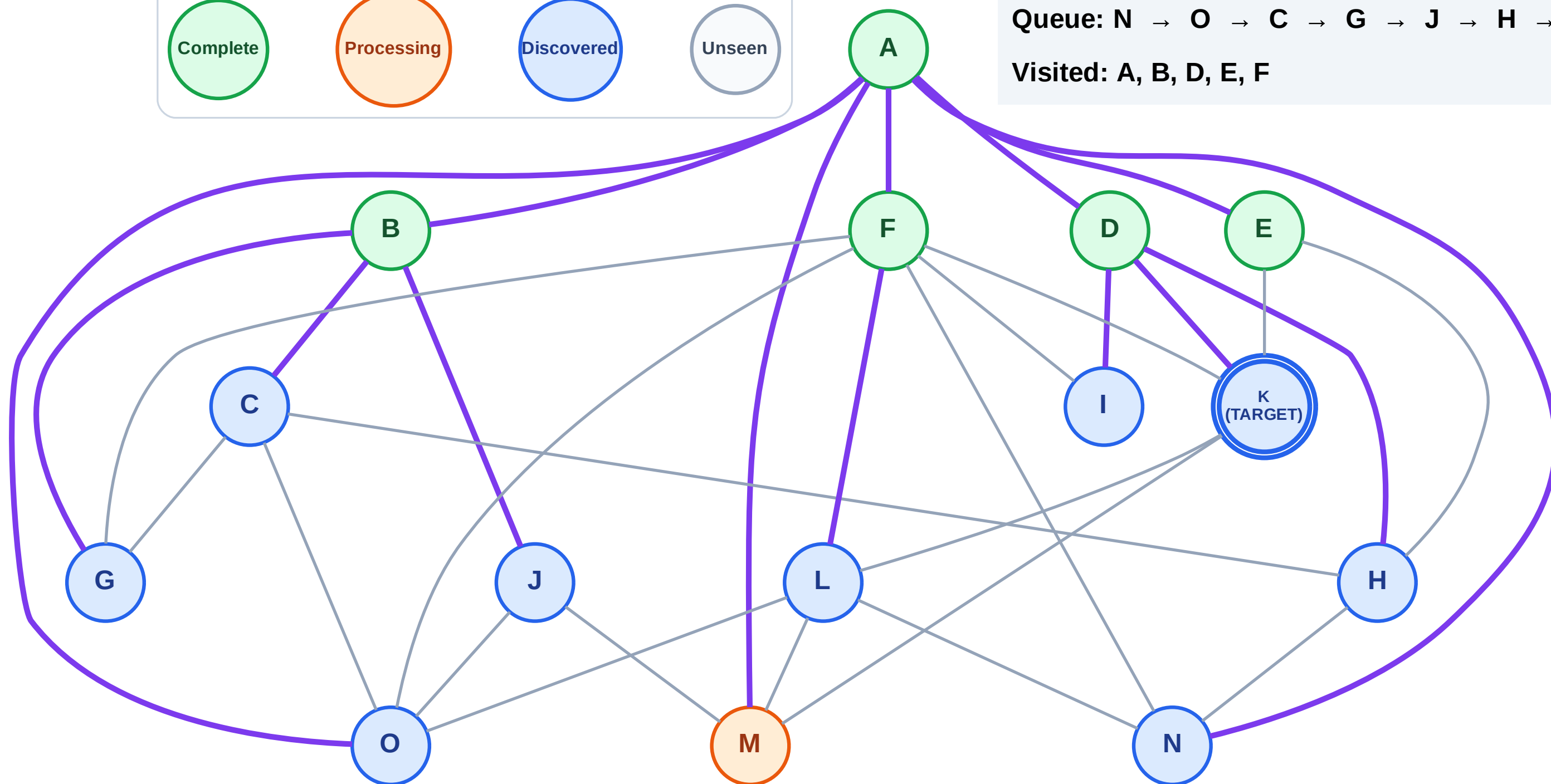
BFS Traversal

Step 12: Process node M



Queue: N → O → C → G → J → H → I → K → L

Visited: A, B, D, E, F



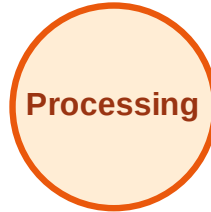
BFS Traversal

Step 13: No new neighbors from M

Legend



Complete



Processing



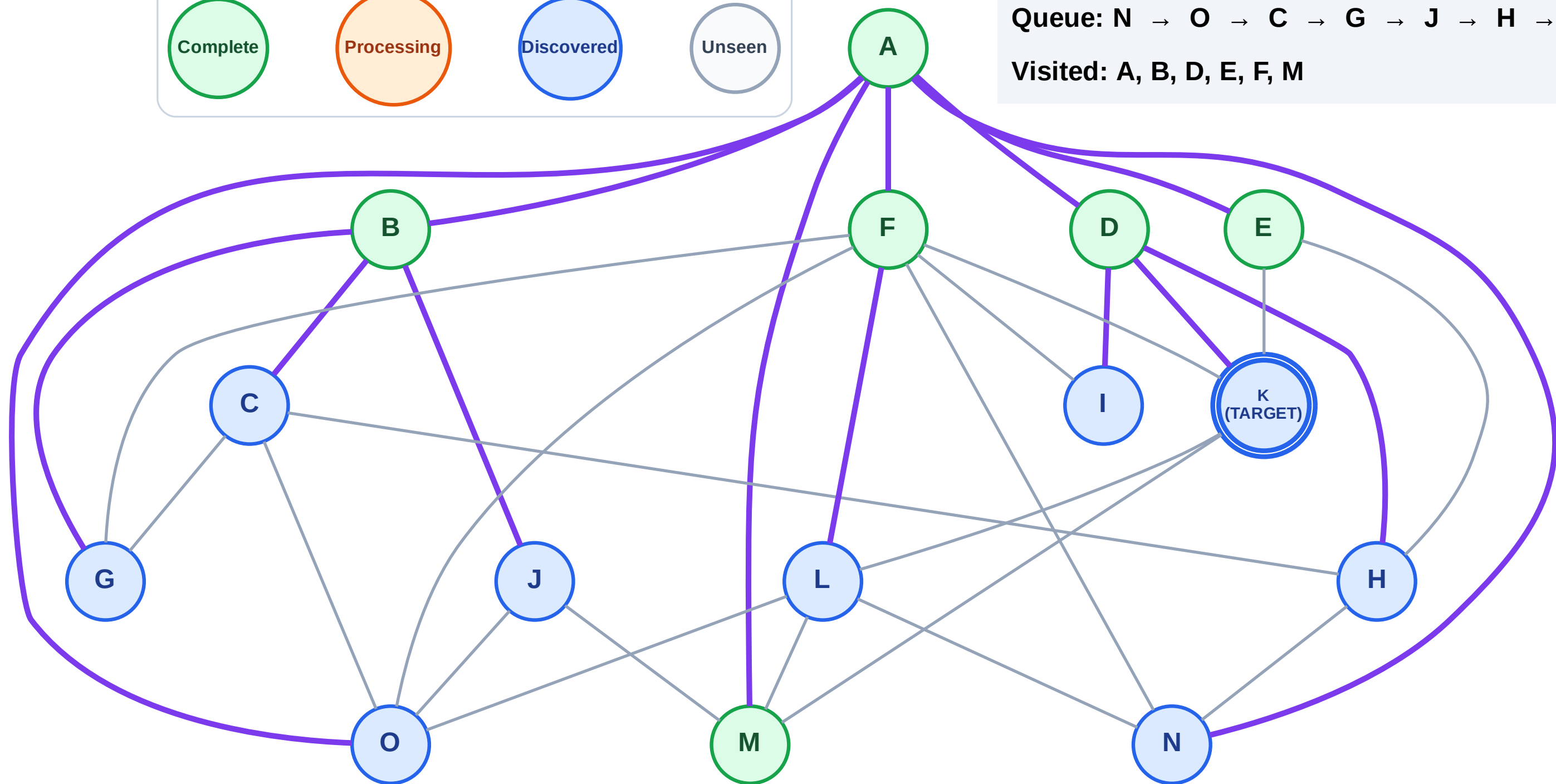
Discovered



Unseen

Queue: N → O → C → G → J → H → I → K → L

Visited: A, B, D, E, F, M



BFS Traversal

Step 14: Process node N

Legend

Complete

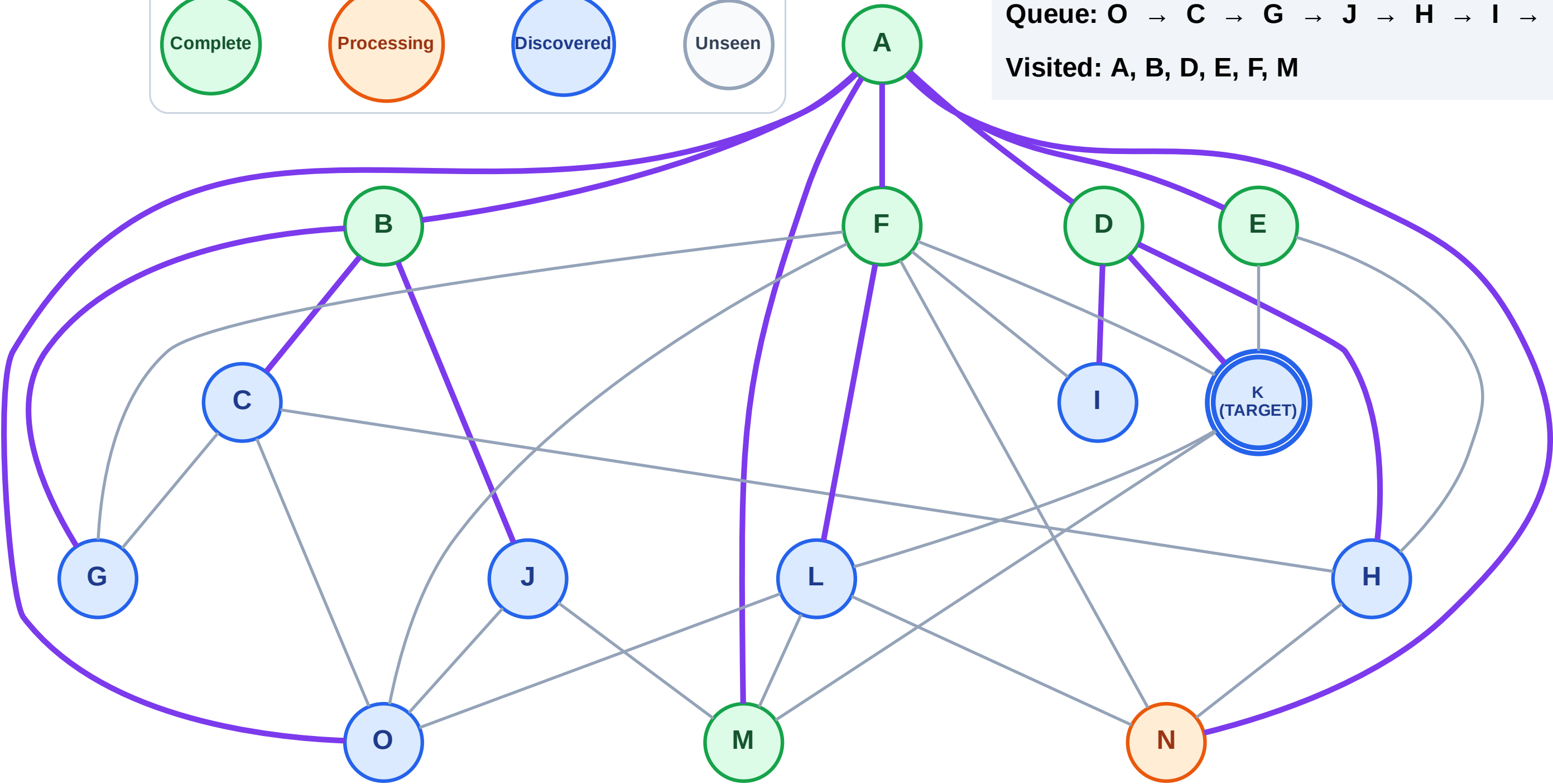
Processing

Discovered

Unseen

Queue: O → C → G → J → H → I → K → L

Visited: A, B, D, E, F, M



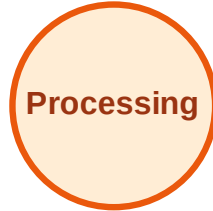
BFS Traversal

Step 15: No new neighbors from N

Legend



Complete



Processing



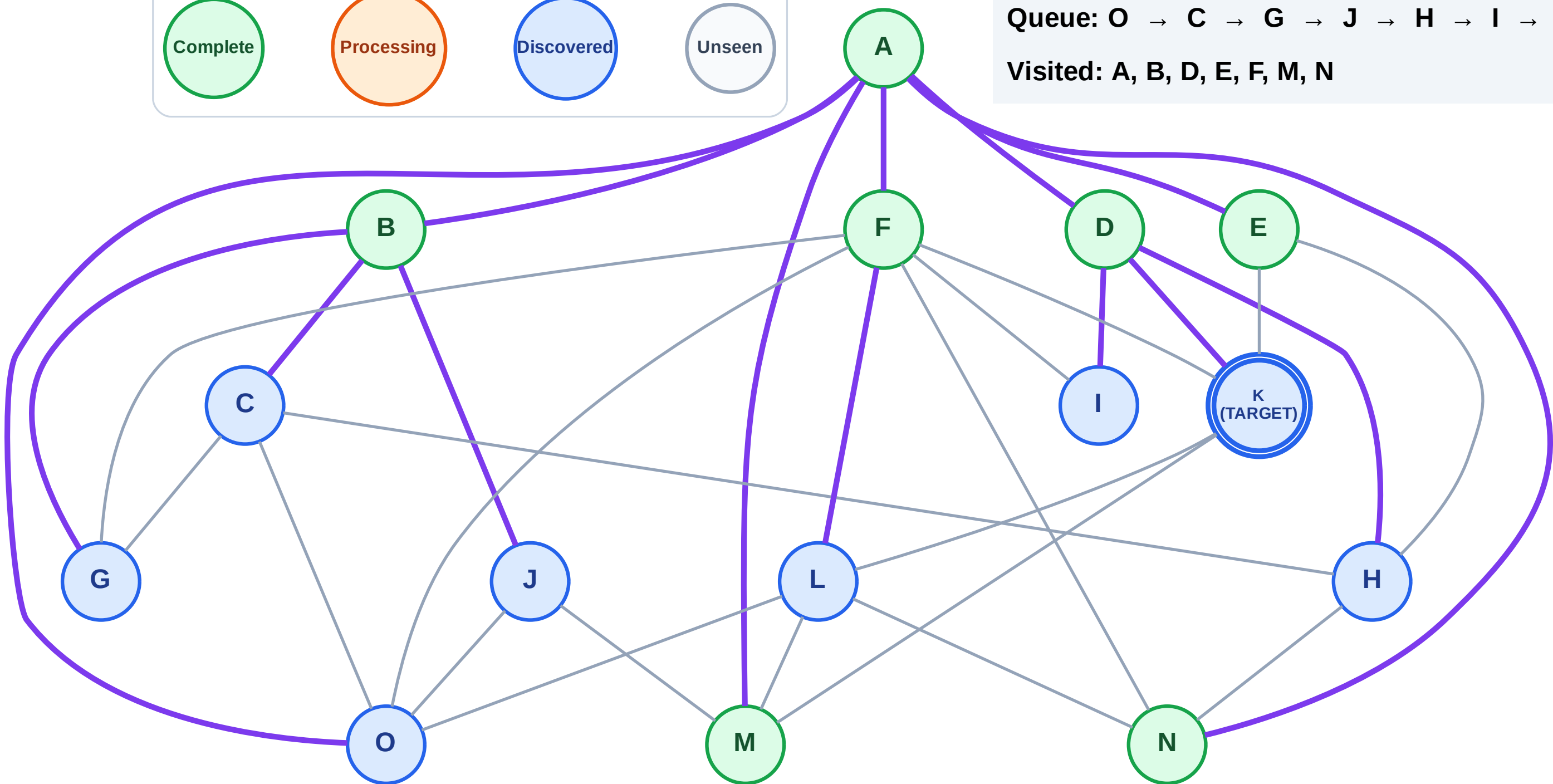
Discovered



Unseen

Queue: O → C → G → J → H → I → K → L

Visited: A, B, D, E, F, M, N



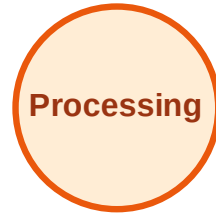
BFS Traversal

Step 16: Process node O

Legend



Complete



Processing



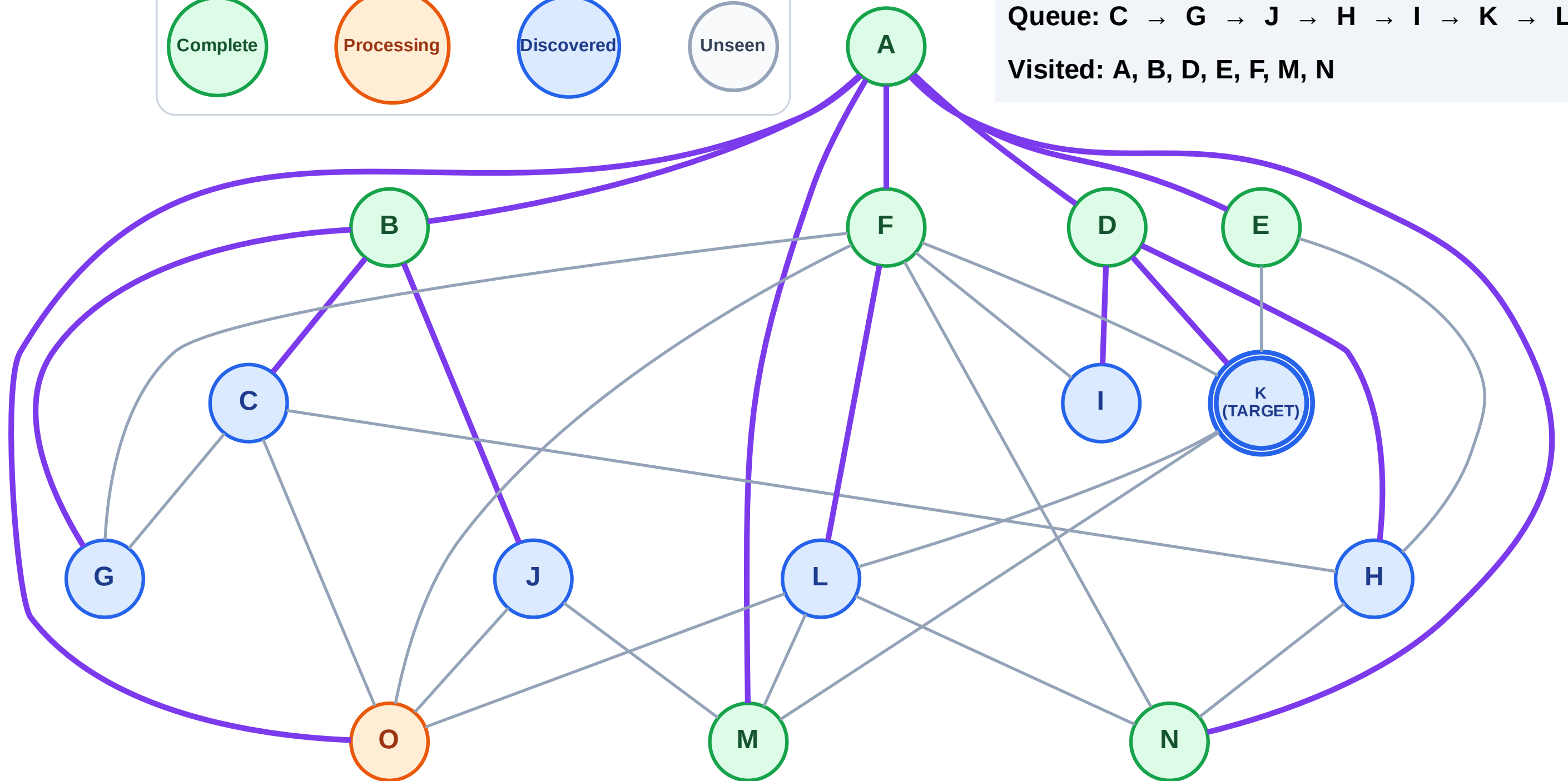
Discovered



Unseen

Queue: C → G → J → H → I → K → L

Visited: A, B, D, E, F, M, N



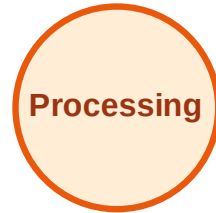
BFS Traversal

Step 17: No new neighbors from O

Legend



Complete



Processing



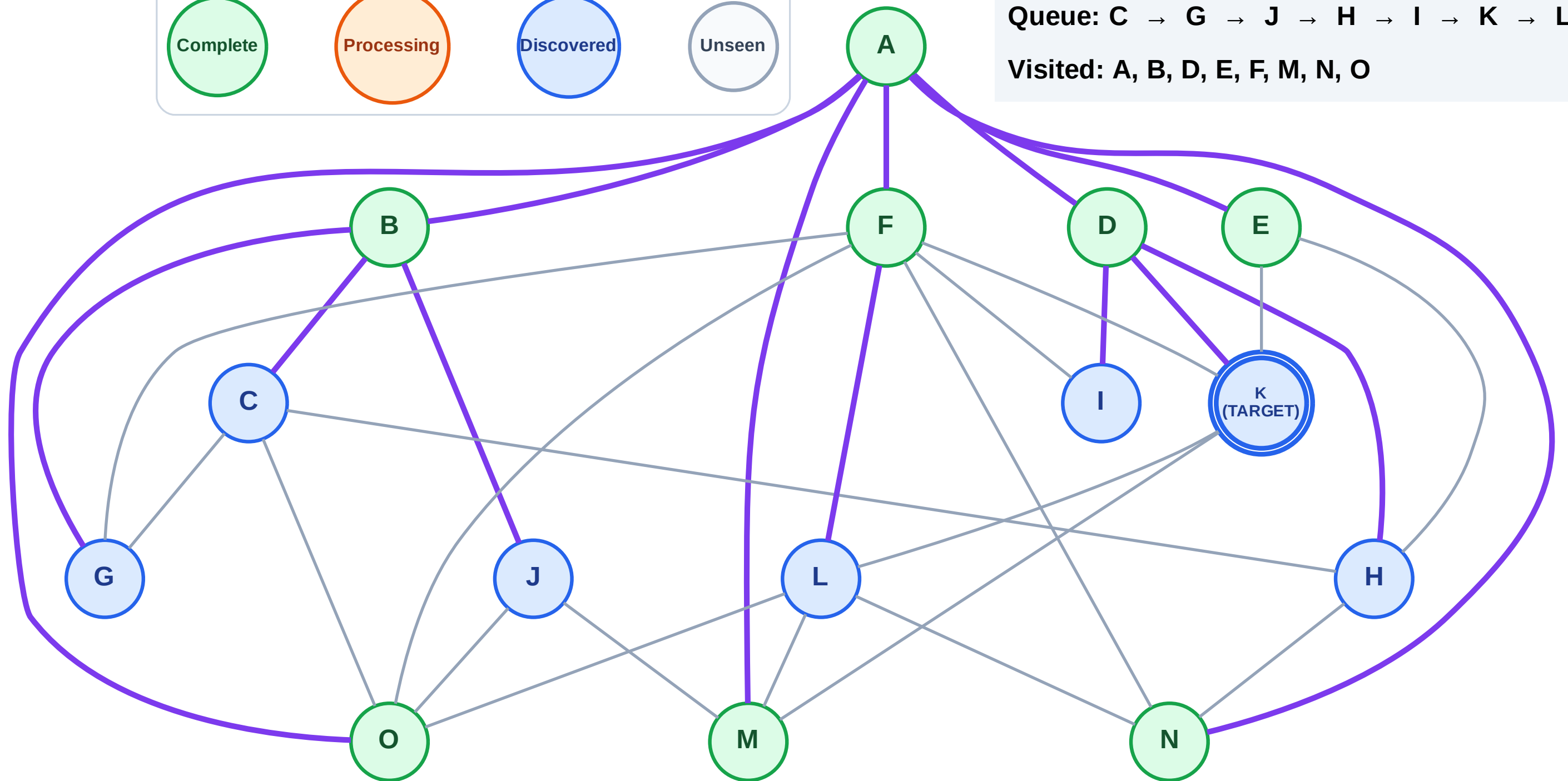
Discovered



Unseen

Queue: C → G → J → H → I → K → L

Visited: A, B, D, E, F, M, N, O



BFS Traversal

Step 18: Process node C

Legend

Complete

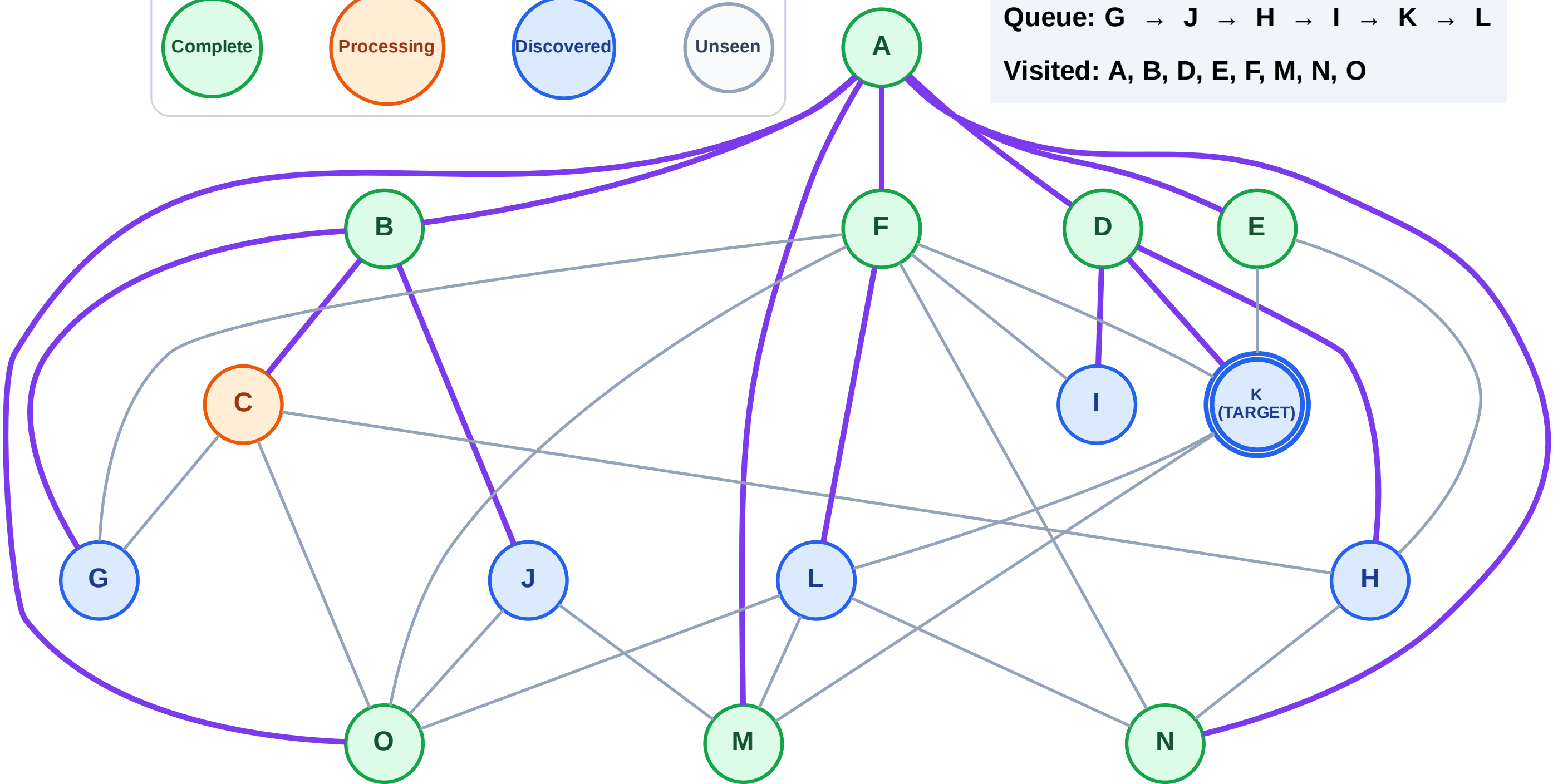
Processing

Discovered

Unseen

Queue: G → J → H → I → K → L

Visited: A, B, D, E, F, M, N, O



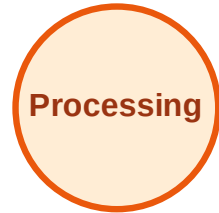
BFS Traversal

Step 19: No new neighbors from C

Legend



Complete



Processing



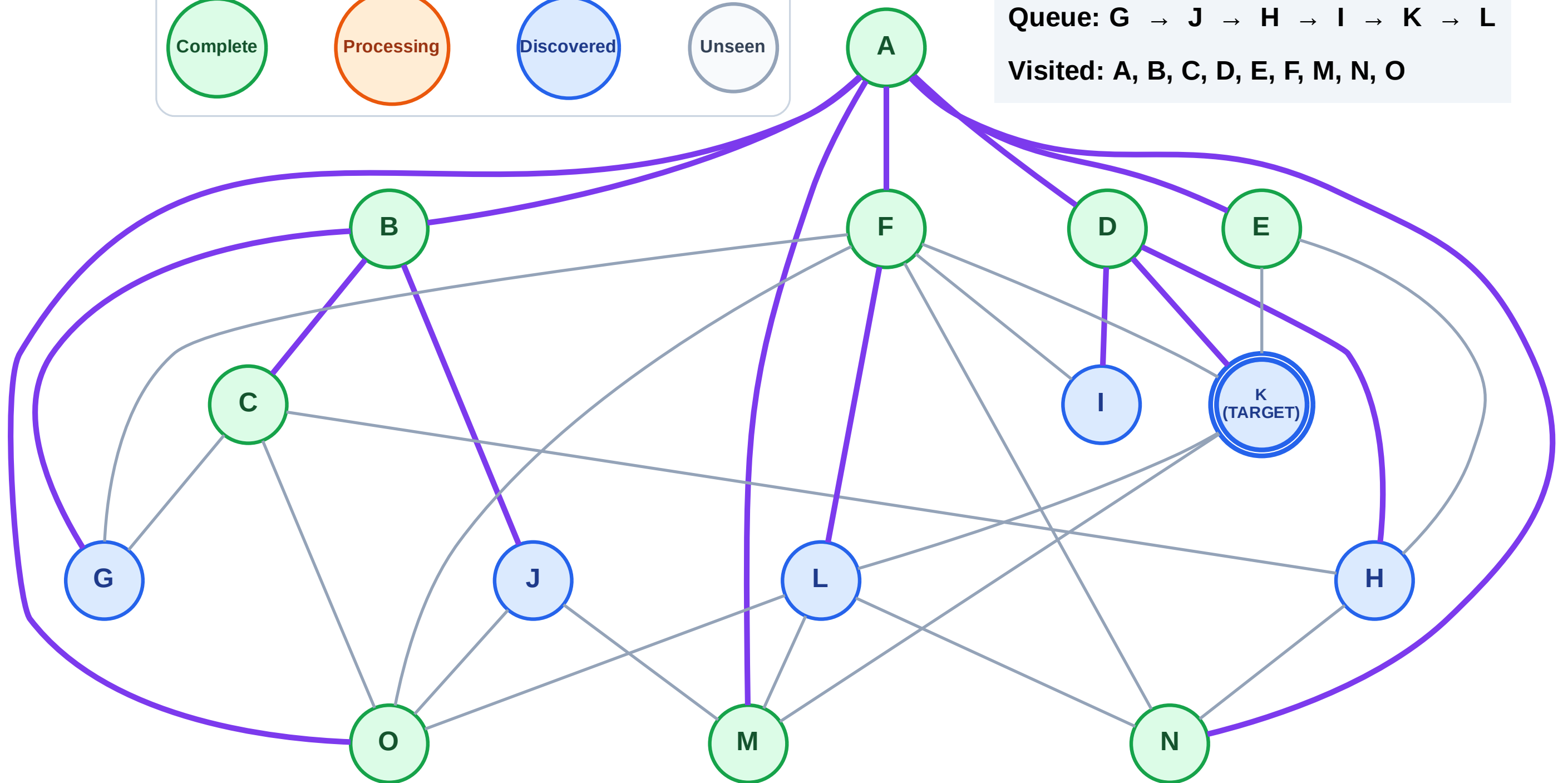
Discovered



Unseen

Queue: G → J → H → I → K → L

Visited: A, B, C, D, E, F, M, N, O



BFS Traversal

Step 20: Process node G

Legend

Complete

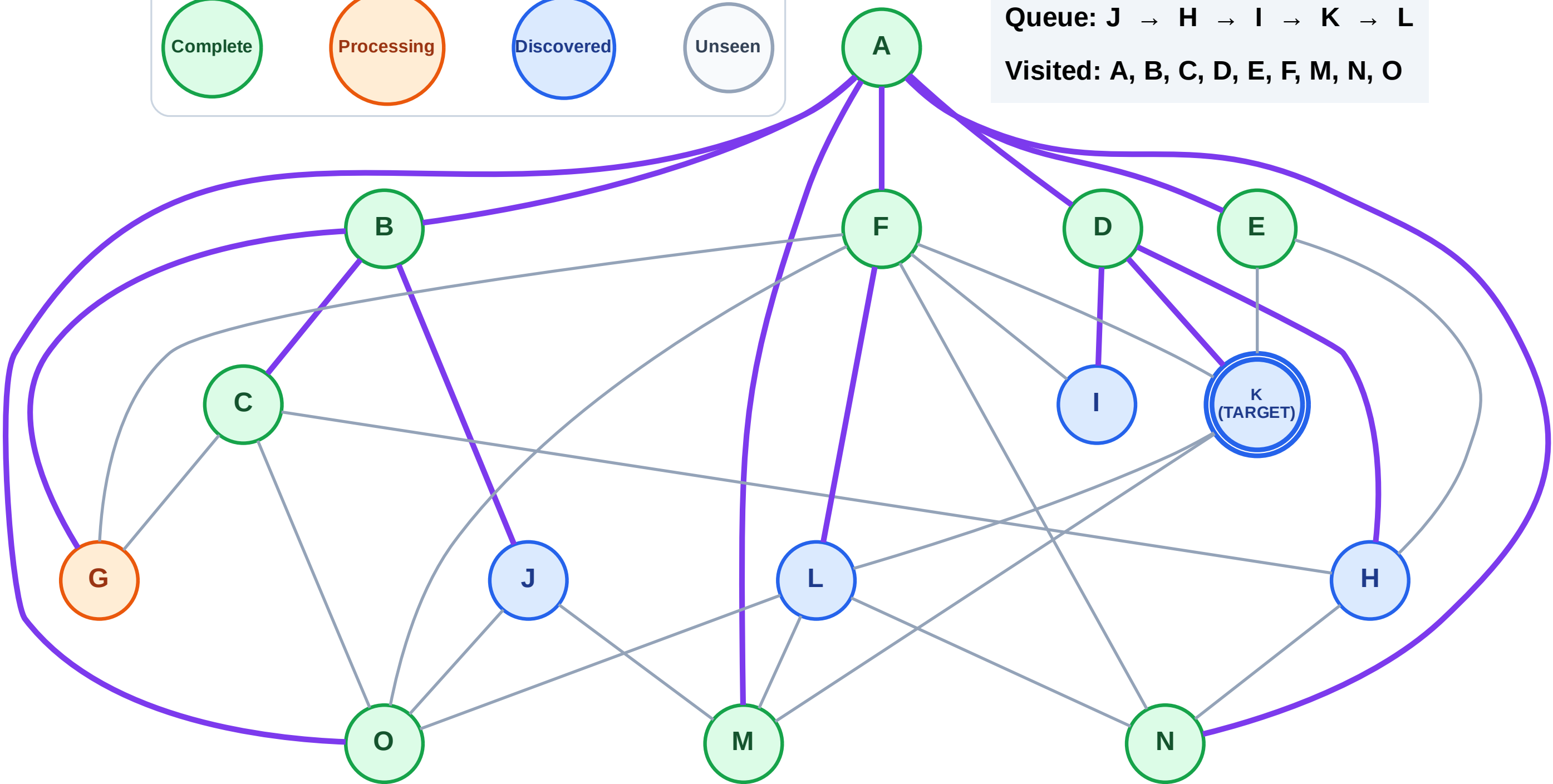
Processing

Discovered

Unseen

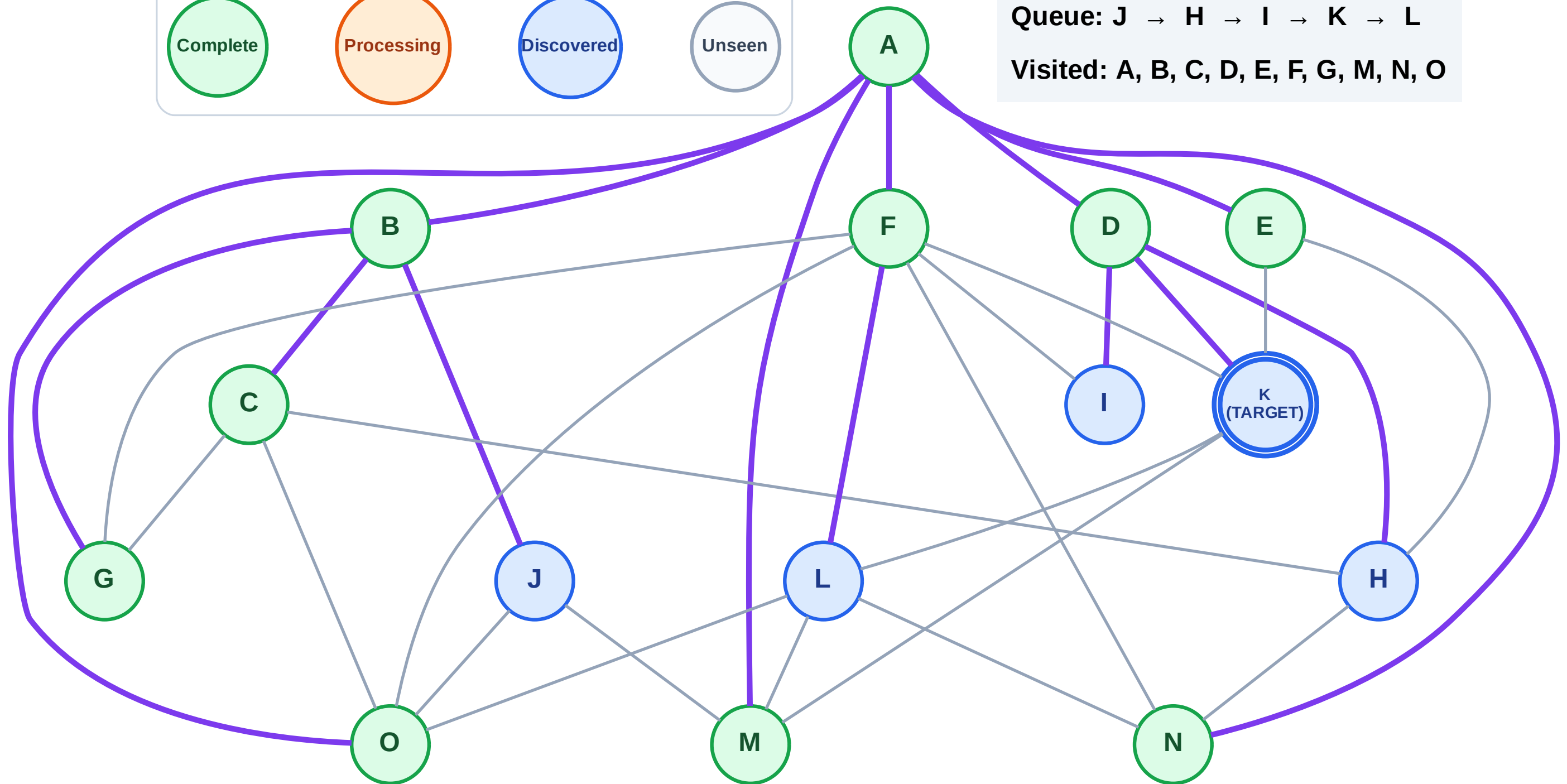
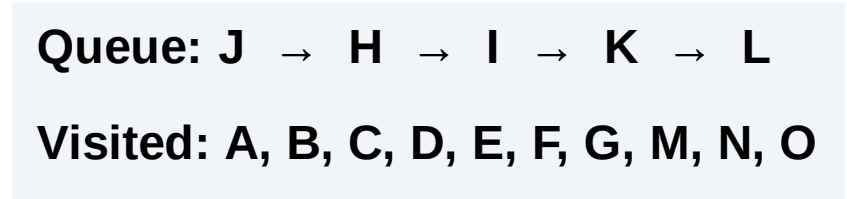
Queue: J → H → I → K → L

Visited: A, B, C, D, E, F, M, N, O



Step 21: No new neighbors from G

Step 21: No new neighbors from G



BFS Traversal

Step 22: Process node J

Legend

Complete

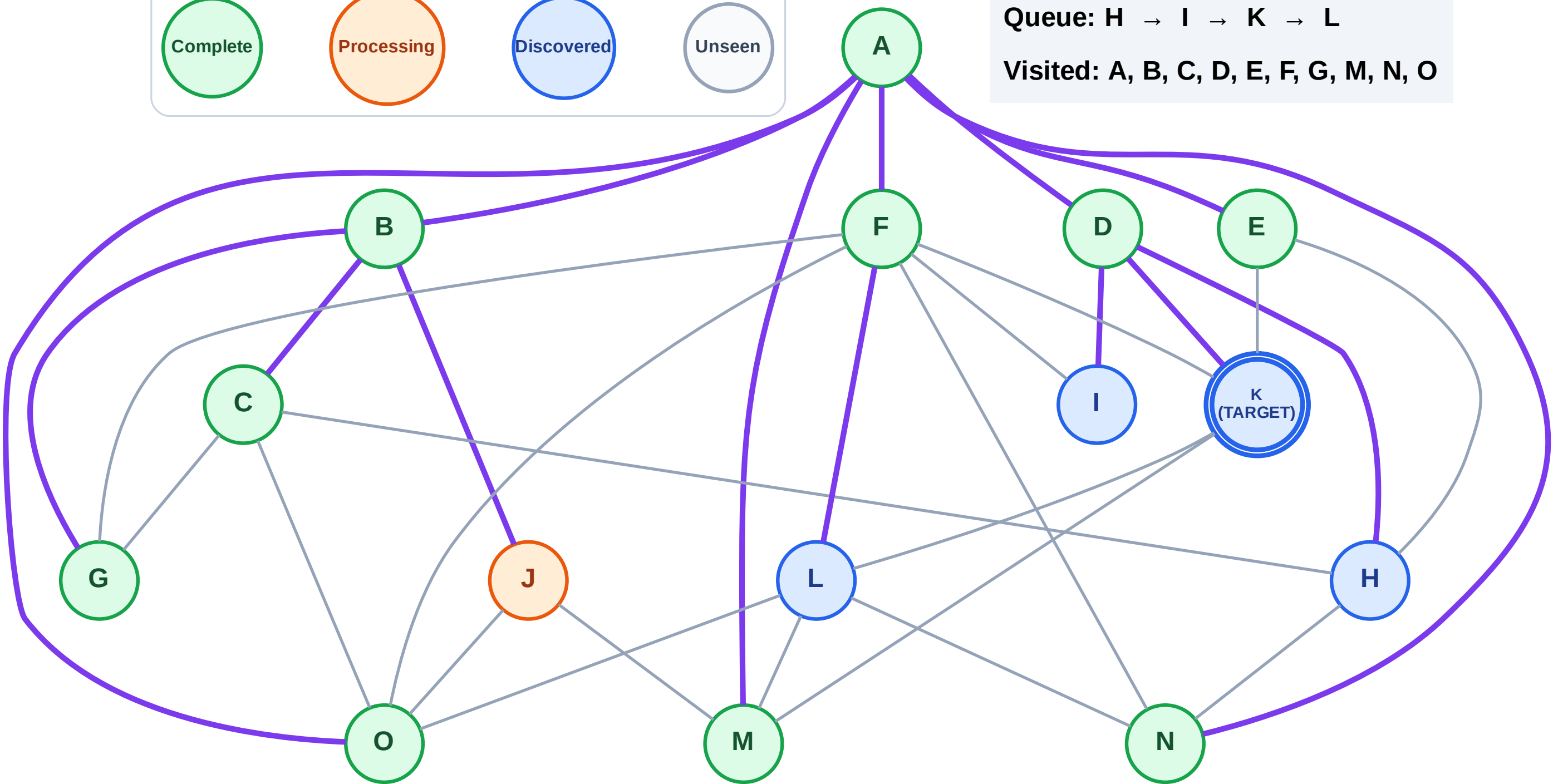
Processing

Discovered

Unseen

Queue: H → I → K → L

Visited: A, B, C, D, E, F, G, M, N, O



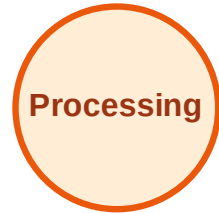
BFS Traversal

Step 23: No new neighbors from J

Legend



Complete



Processing



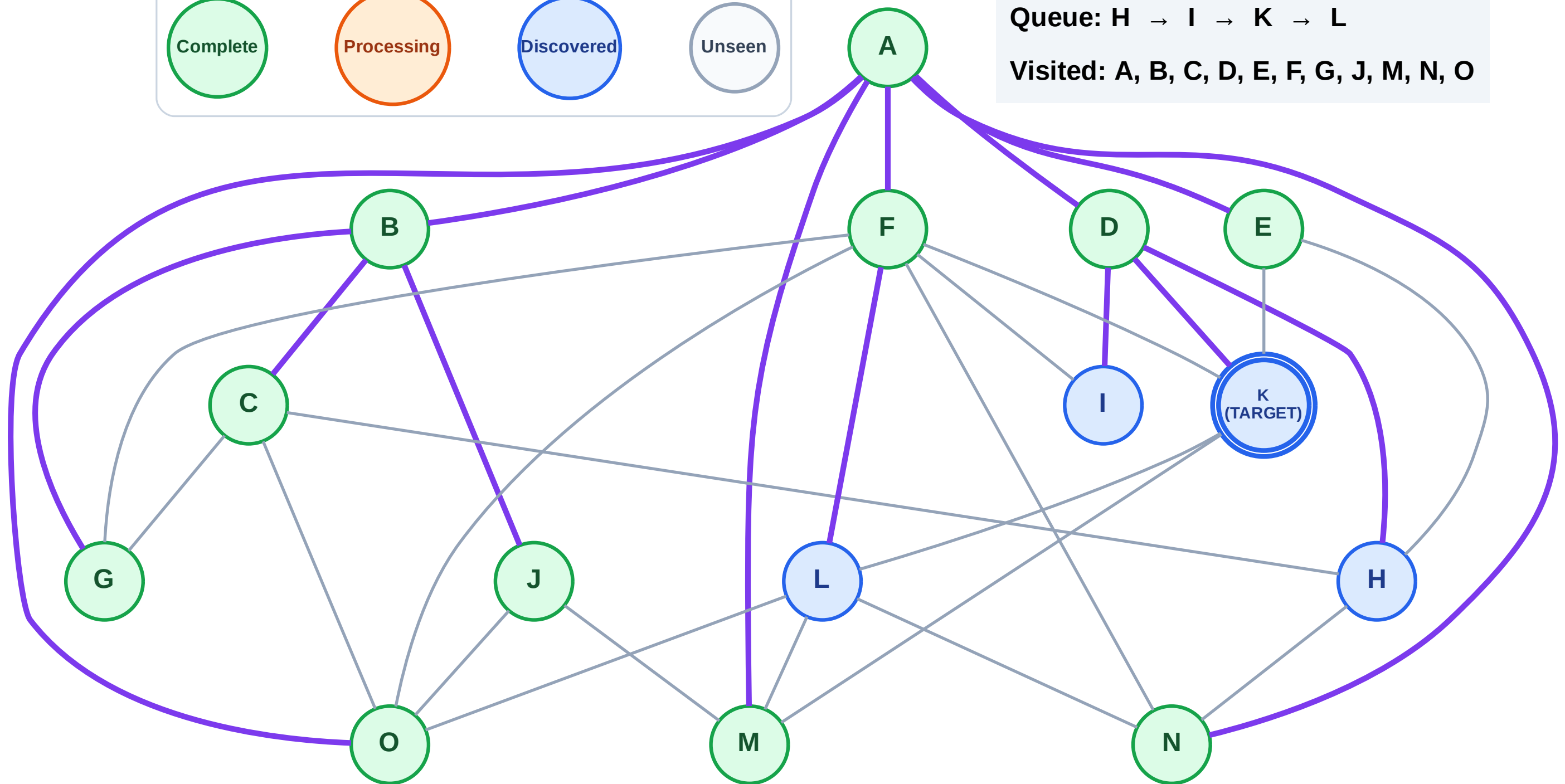
Discovered



Unseen

Queue: H → I → K → L

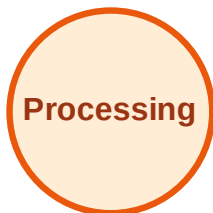
Visited: A, B, C, D, E, F, G, J, M, N, O



BFS Traversal

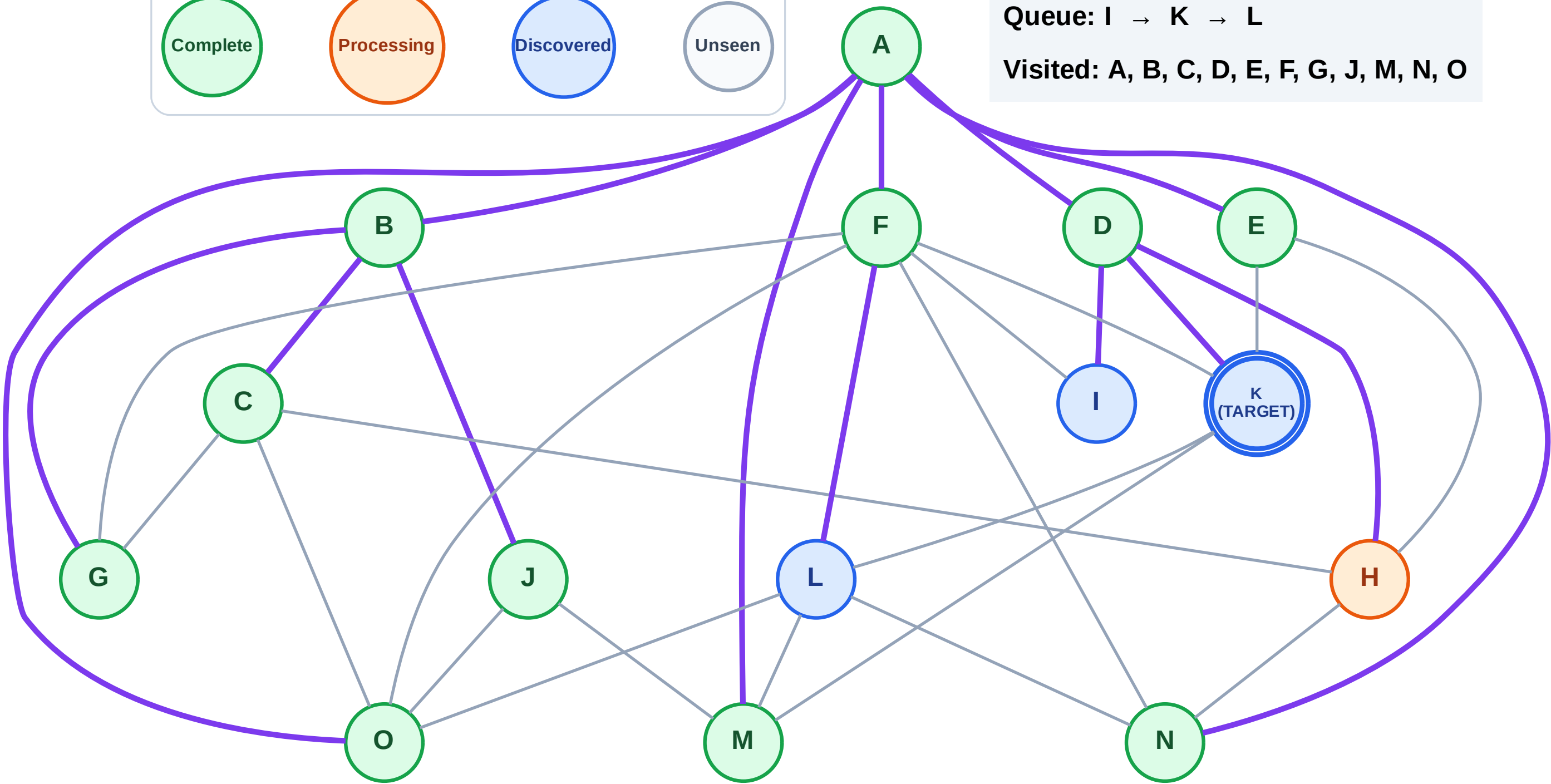
Step 24: Process node H

Legend



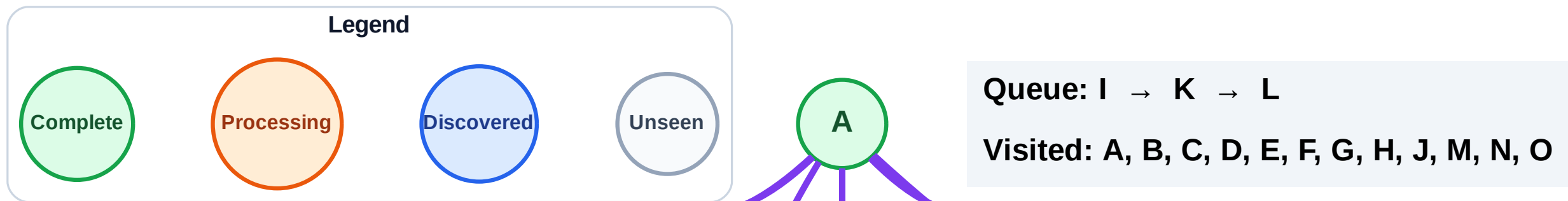
Queue: I → K → L

Visited: A, B, C, D, E, F, G, J, M, N, O



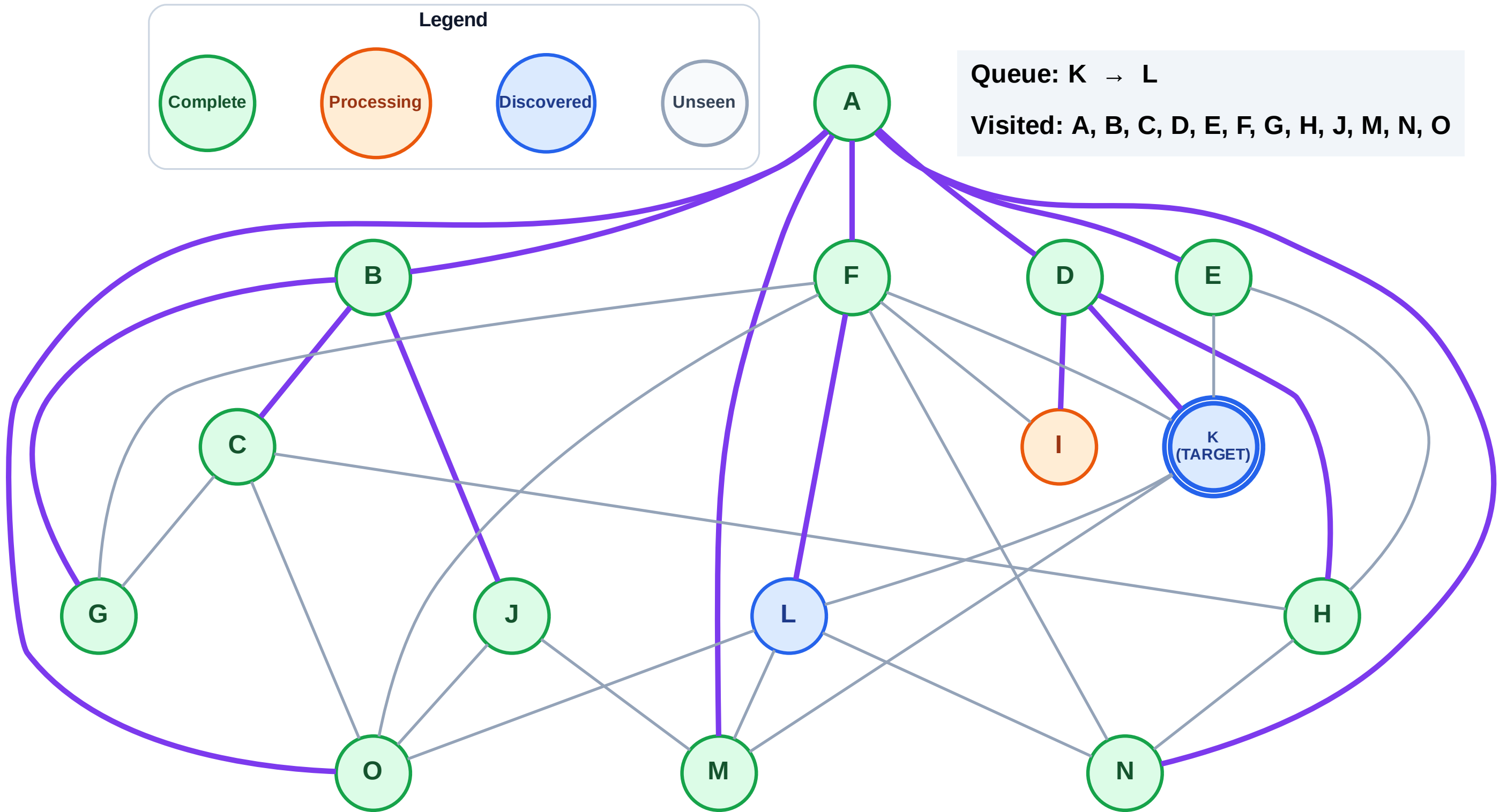
Step 25: No new neighbors from H

Step 25: No new neighbors from H



Step 26: Process node I

Page 10 of 10



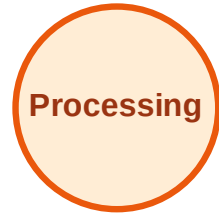
BFS Traversal

Step 27: No new neighbors from I

Legend



Complete



Processing



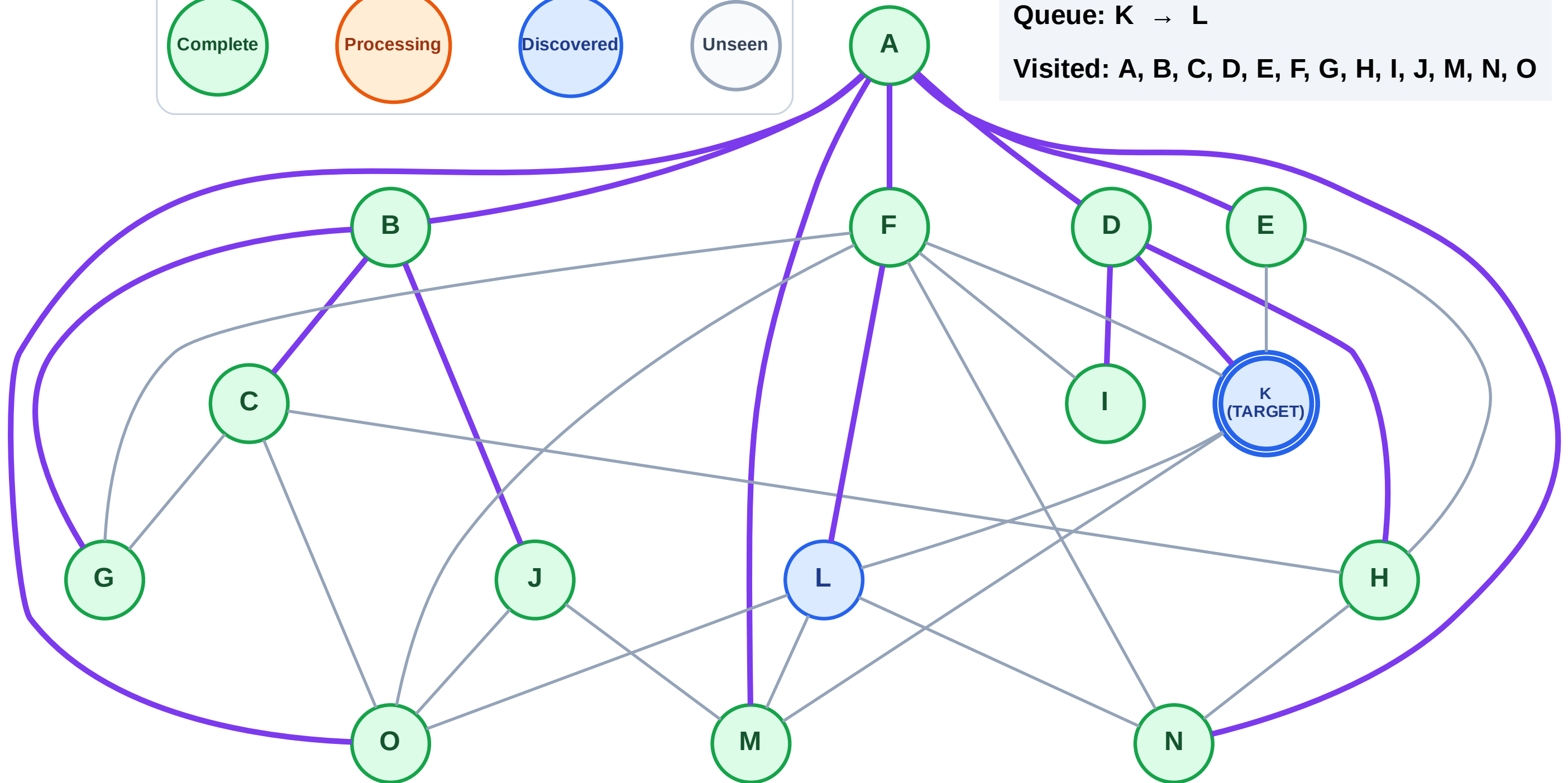
Discovered



Unseen

Queue: K → L

Visited: A, B, C, D, E, F, G, H, I, J, M, N, O



BFS Traversal

Step 28: Process node K

Legend

Complete

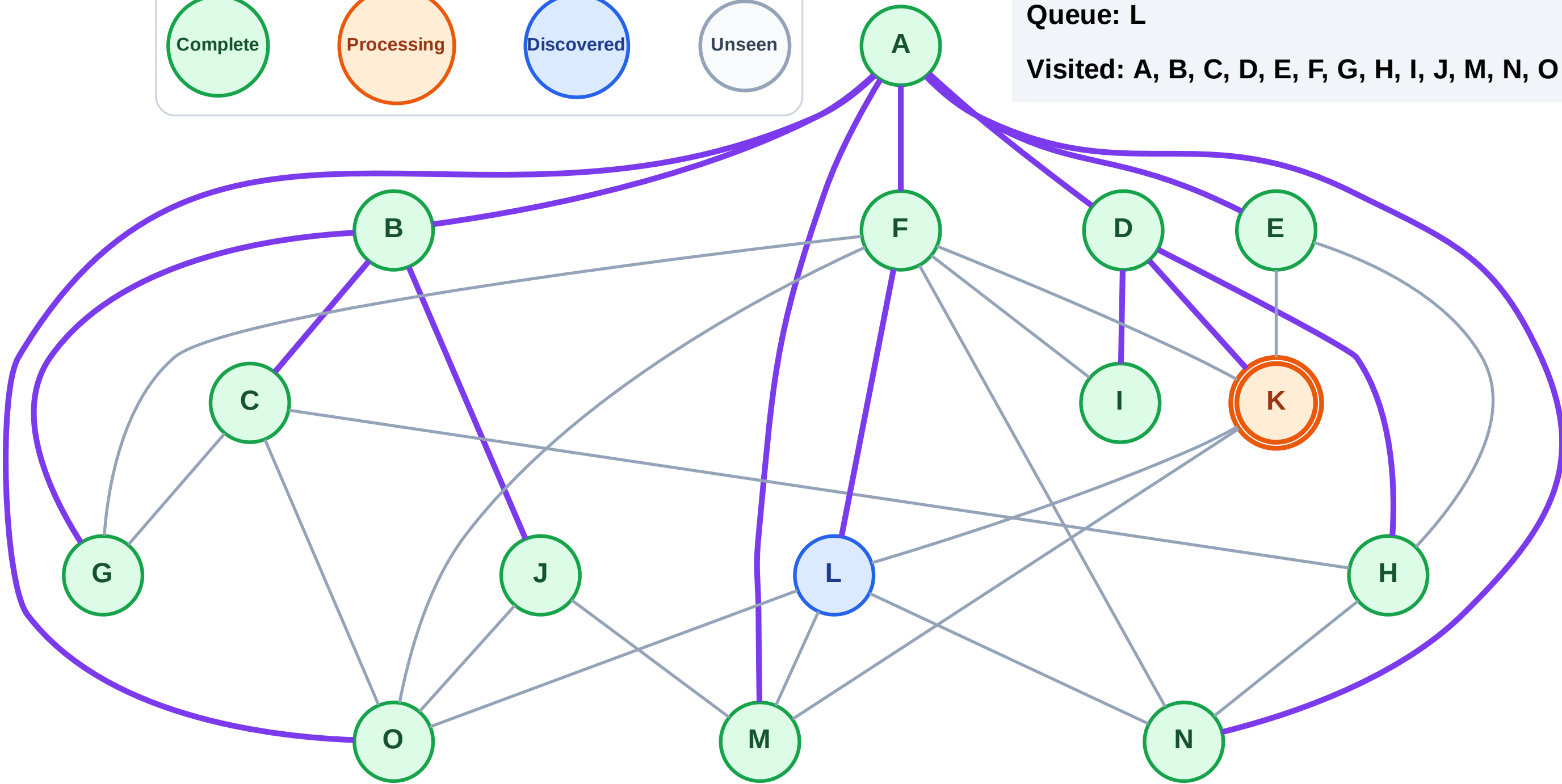
Processing

Discovered

Unseen

Queue: L

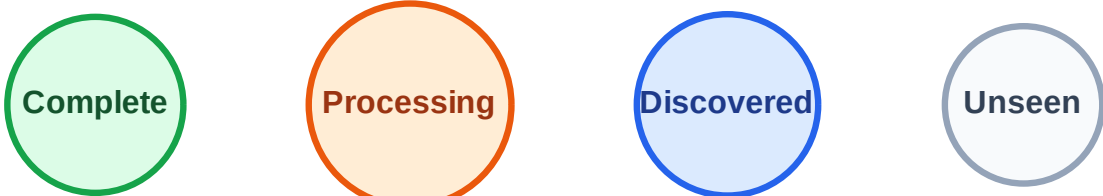
Visited: A, B, C, D, E, F, G, H, I, J, M, N, O



BFS Traversal

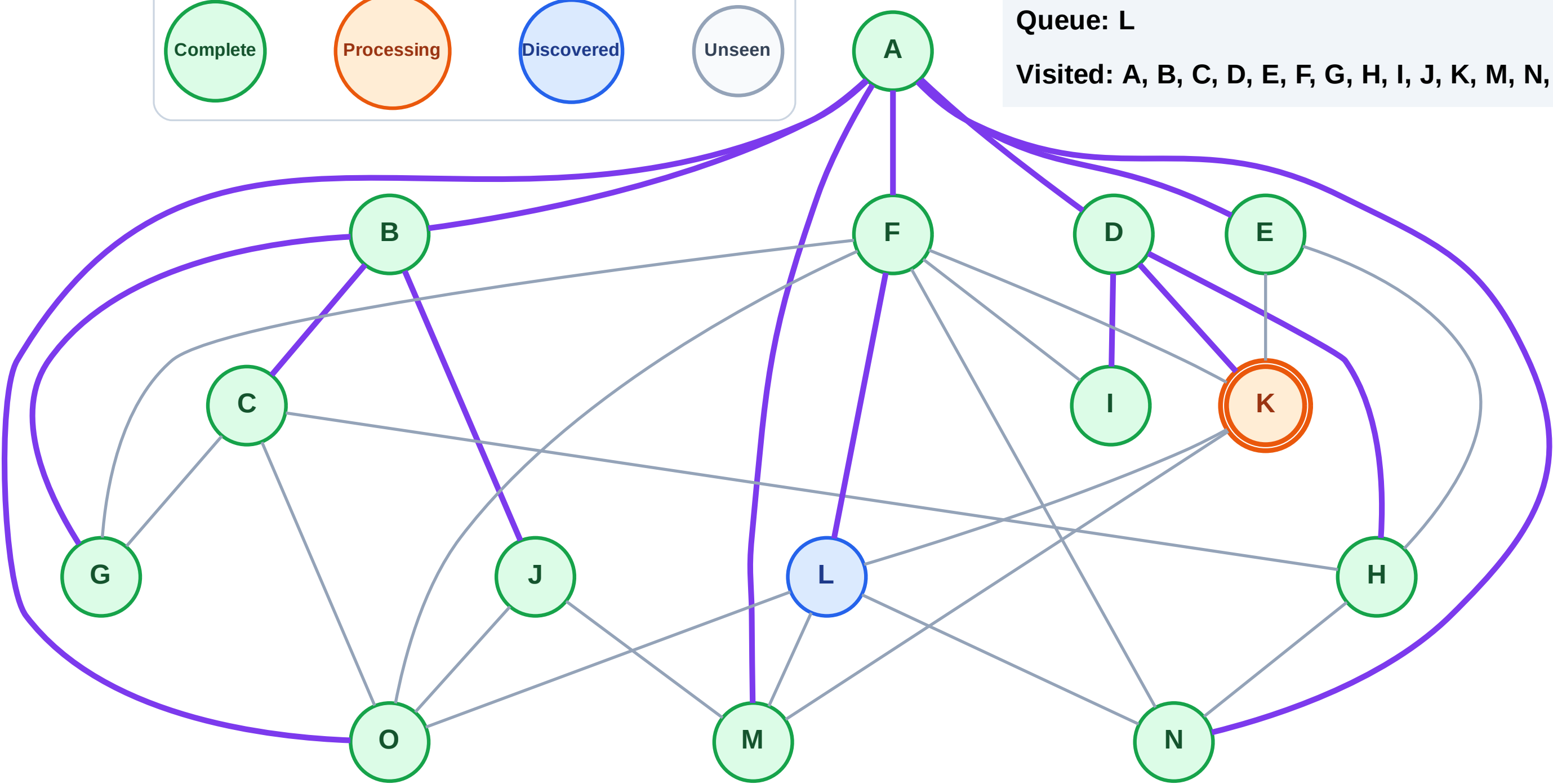
Step 29: TARGET NODE K REACHED! Path (2 hops): A → D → K

Legend



Queue: L

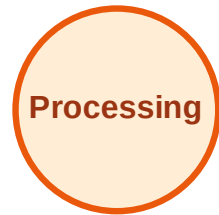
Visited: A, B, C, D, E, F, G, H, I, J, K, M, N, O



DFS Traversal

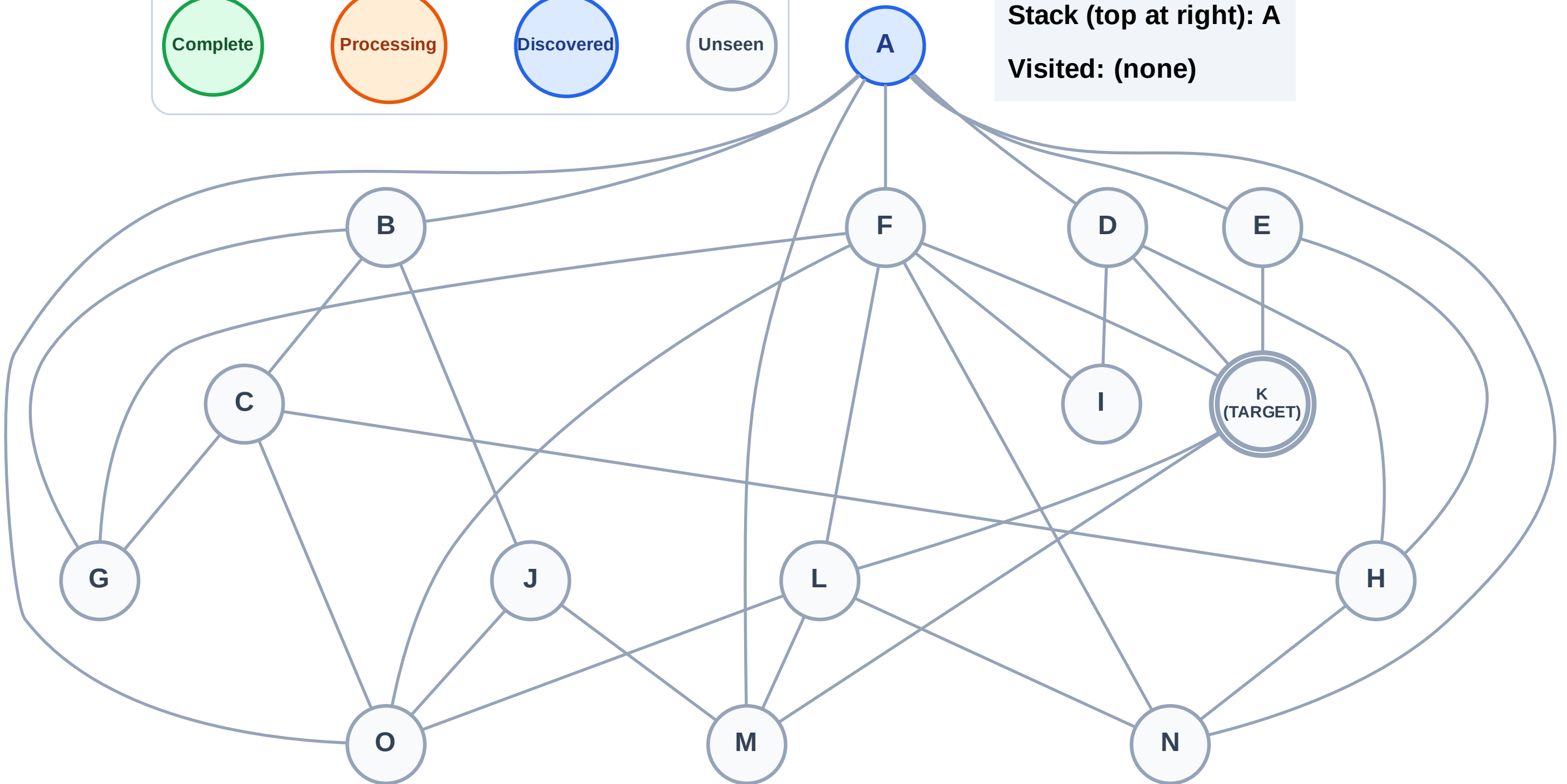
Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

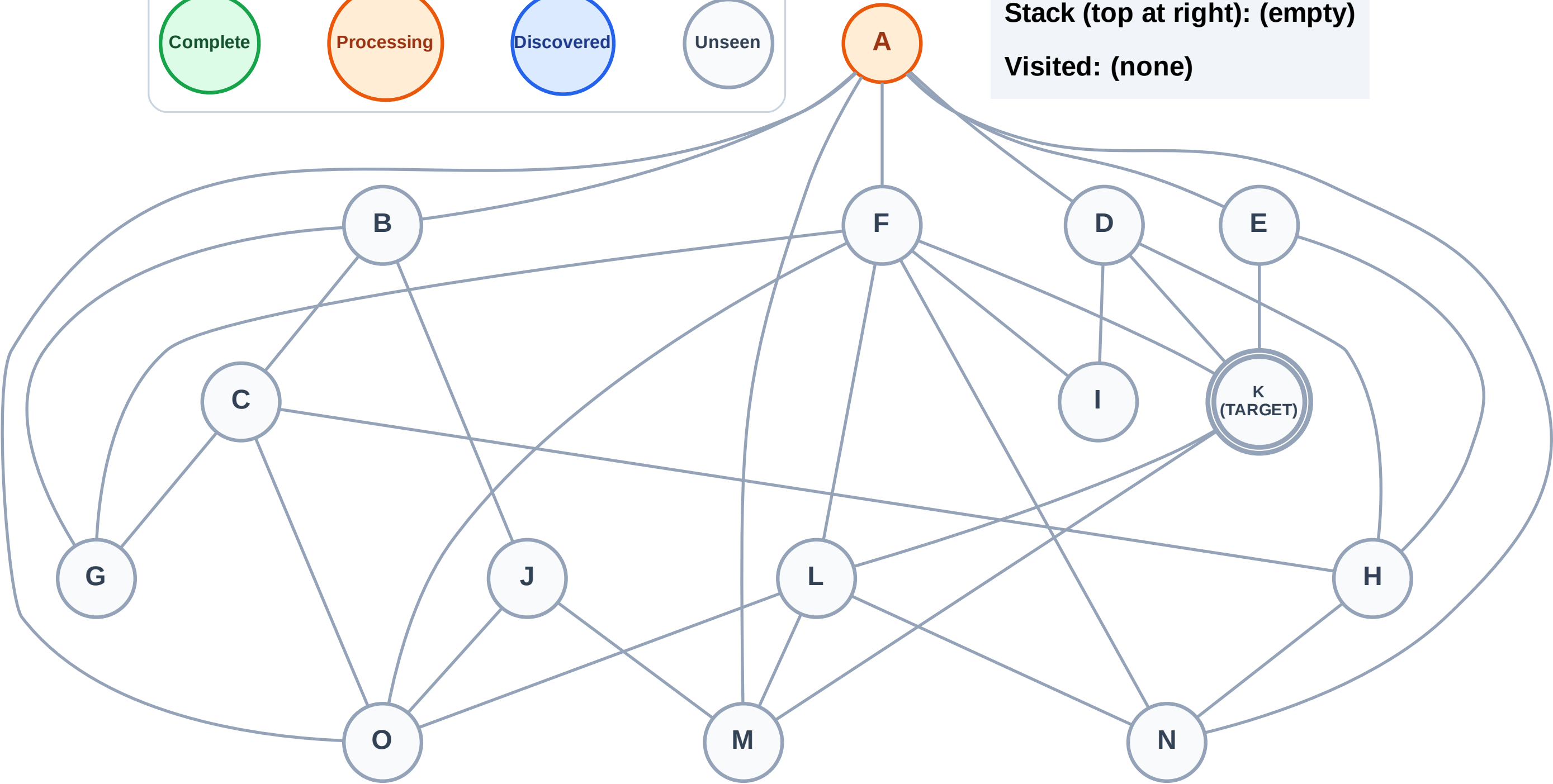
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

Step 3: Discover O, N, M, F, E, D, B from A

Legend

Complete

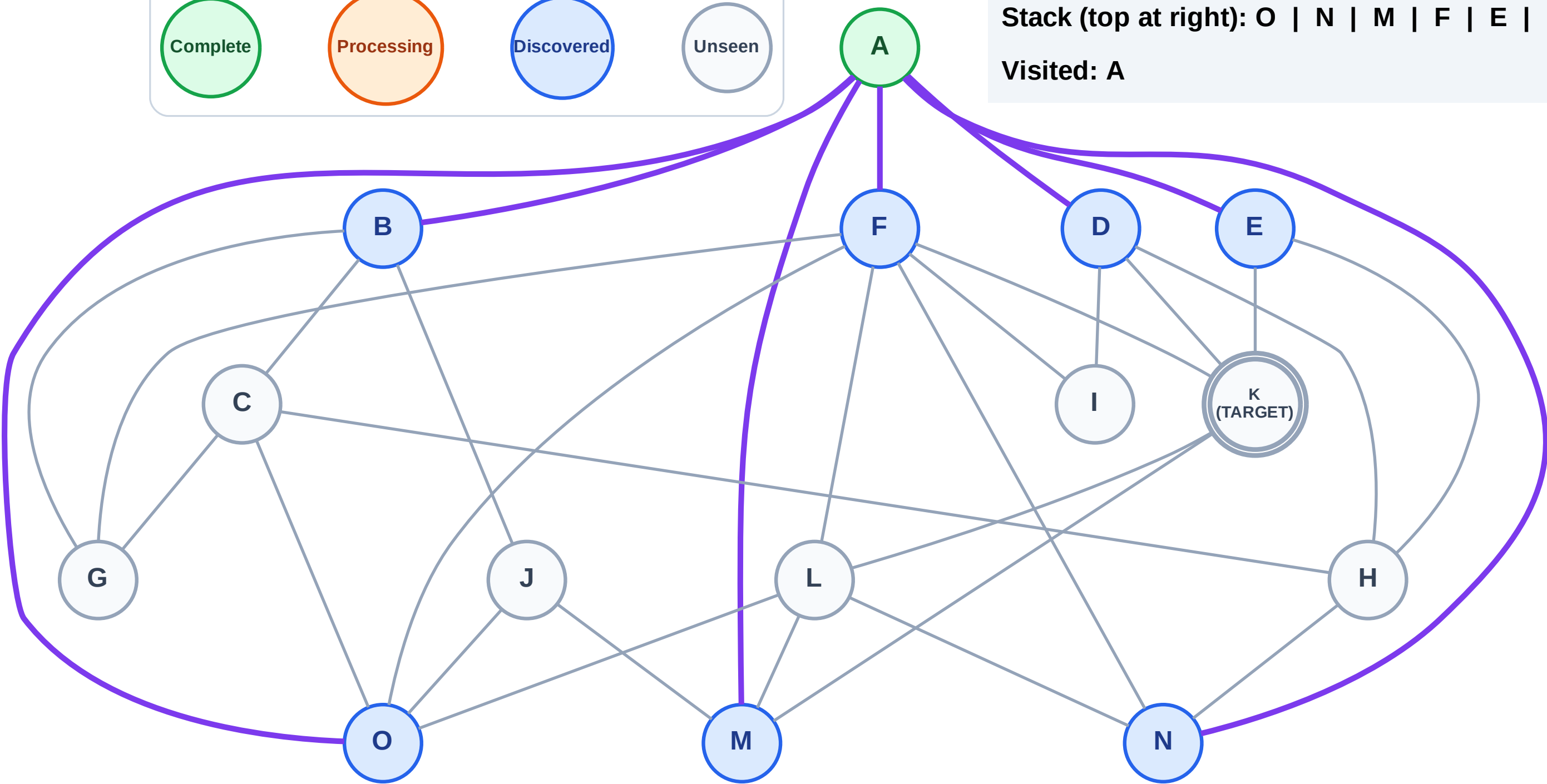
Processing

Discovered

Unseen

Stack (top at right): O | N | M | F | E | D | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

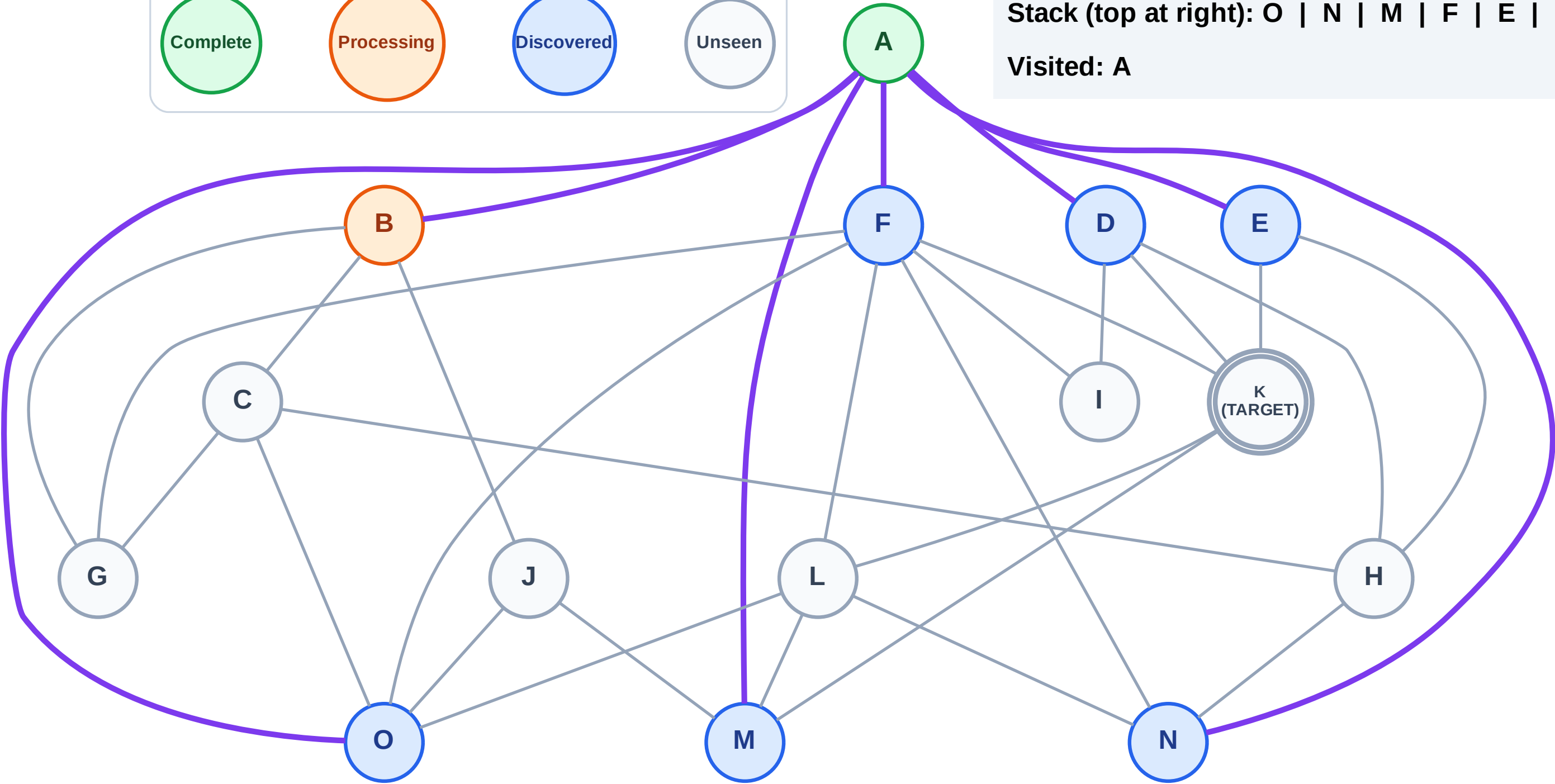
Processing

Discovered

Unseen

Stack (top at right): O | N | M | F | E | D

Visited: A



DFS Traversal

Step 5: Discover J, G, C from B

Legend

Complete

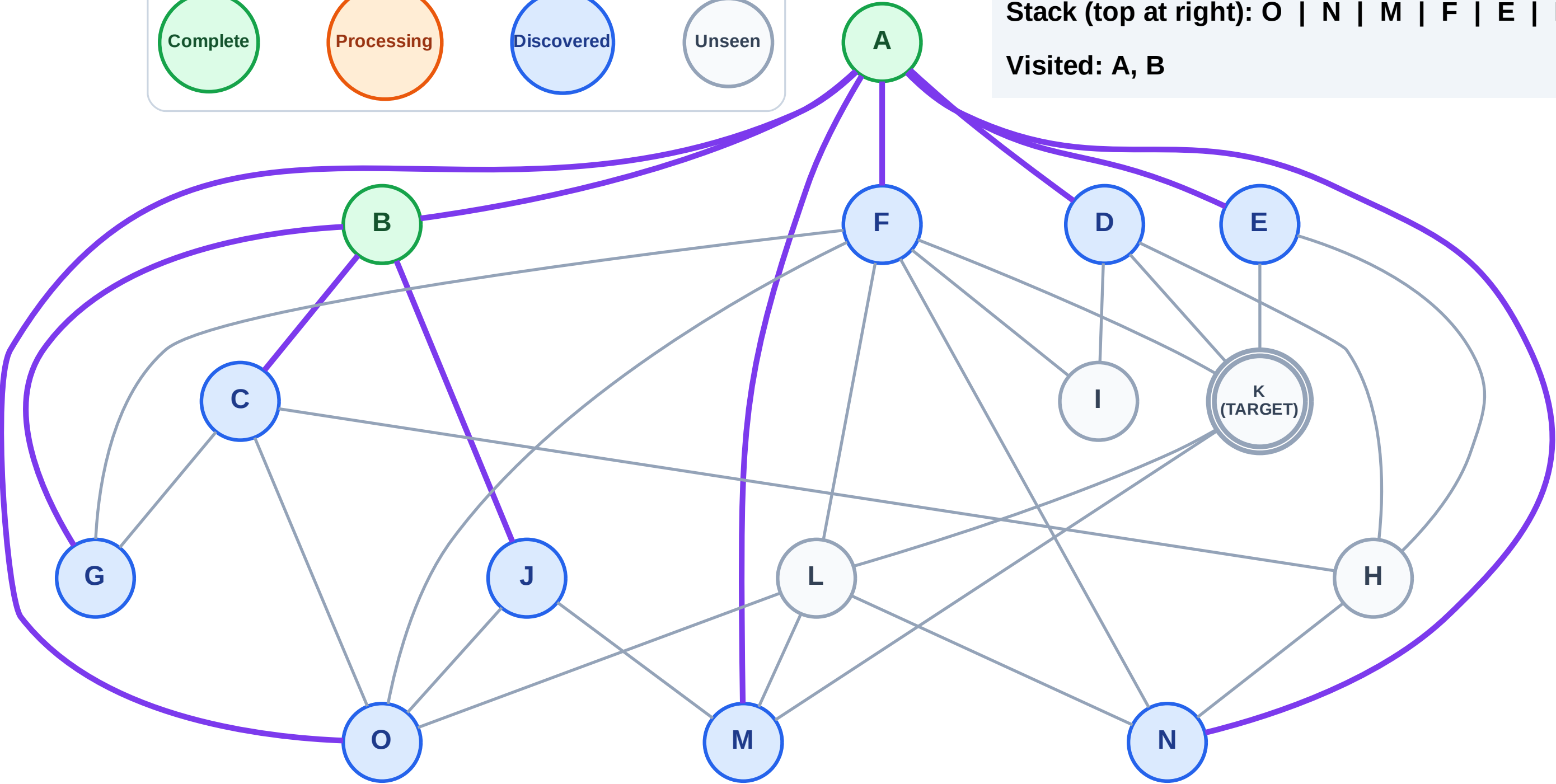
Processing

Discovered

Unseen

Stack (top at right): O | N | M | F | E | D | J | G | C

Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

Complete

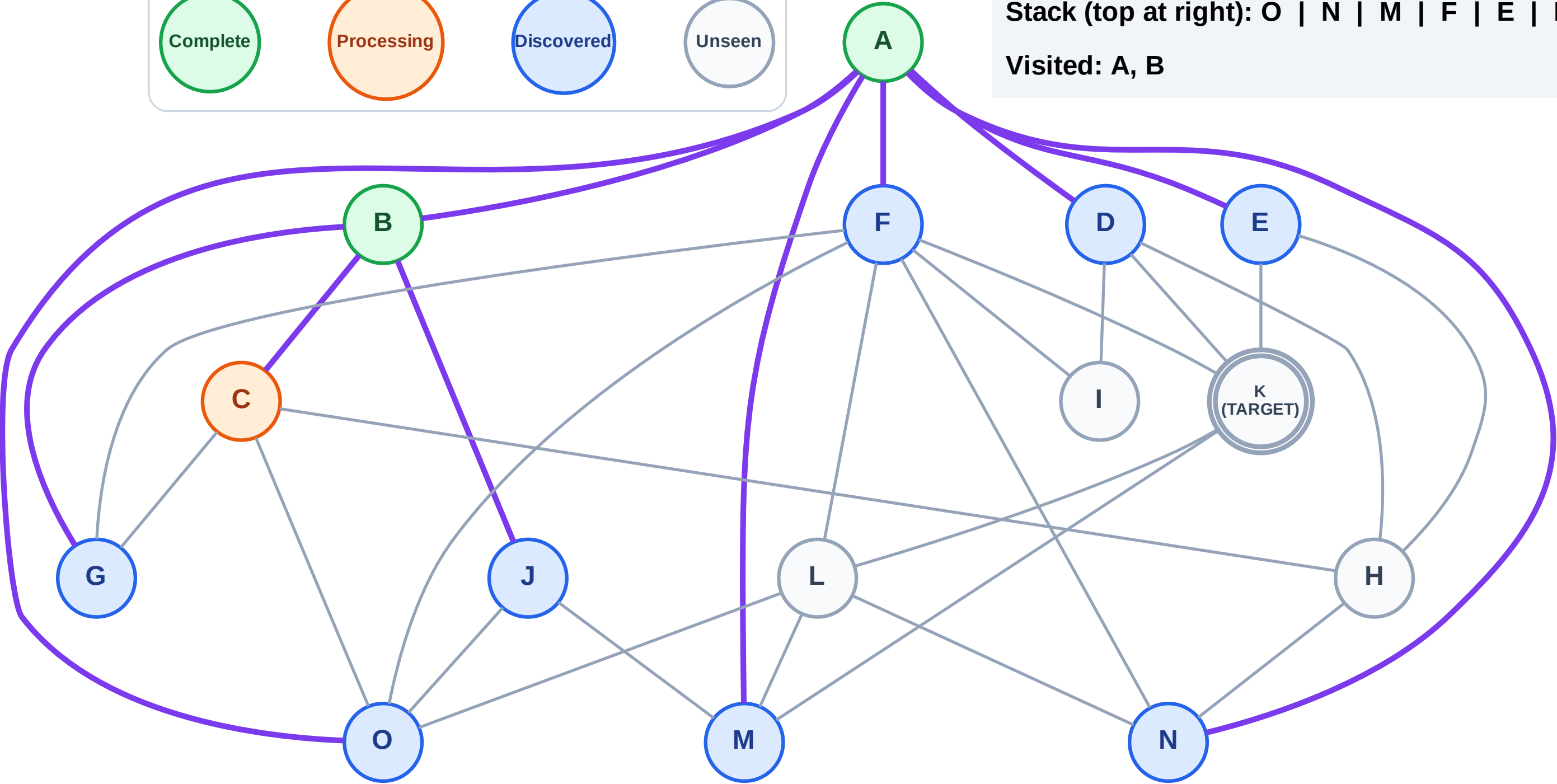
Processing

Discovered

Unseen

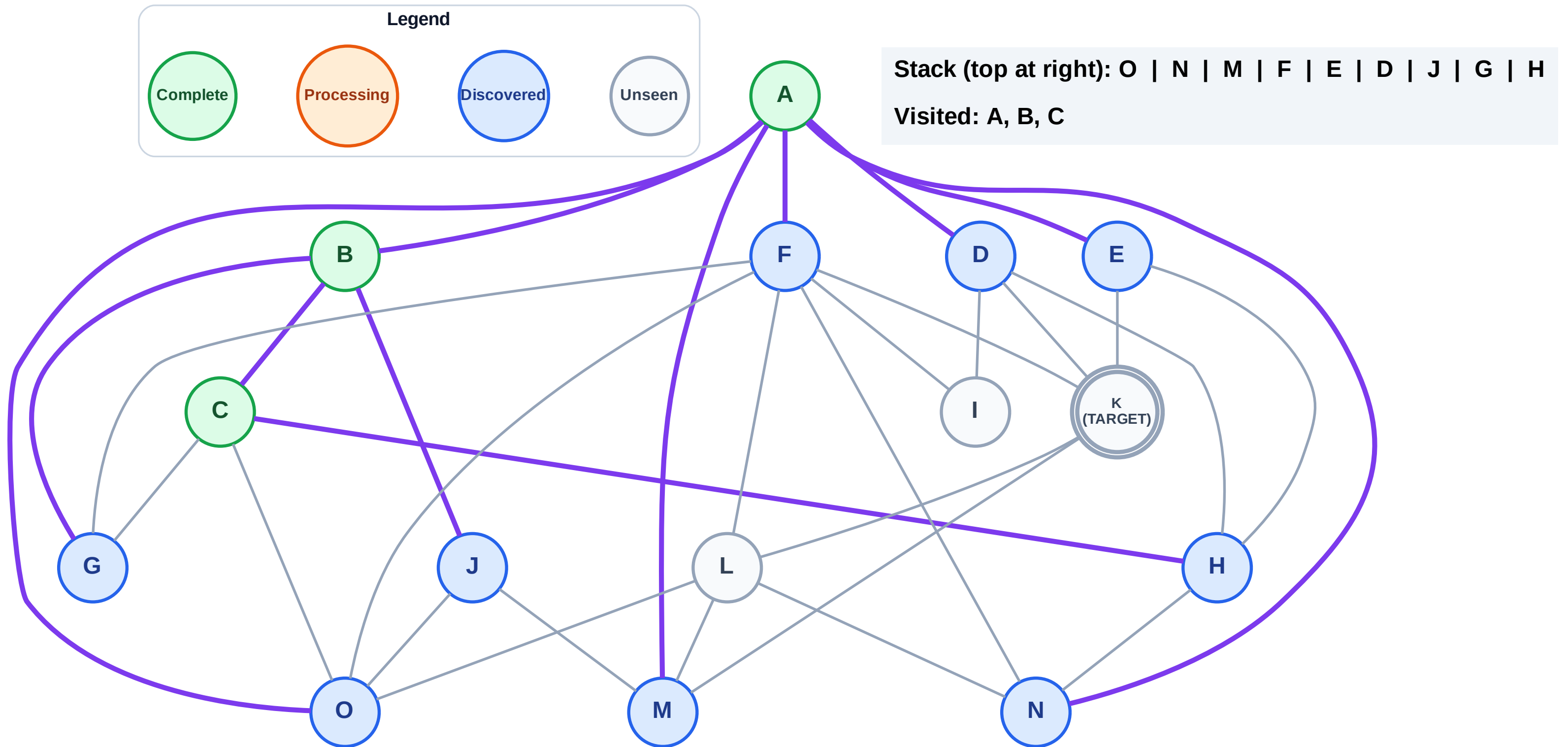
Stack (top at right): O | N | M | F | E | D | J | G

Visited: A, B



Step 7: Discover H from C

Step 7: Discover H from C



DFS Traversal

Step 8: Process node H

Legend

Complete

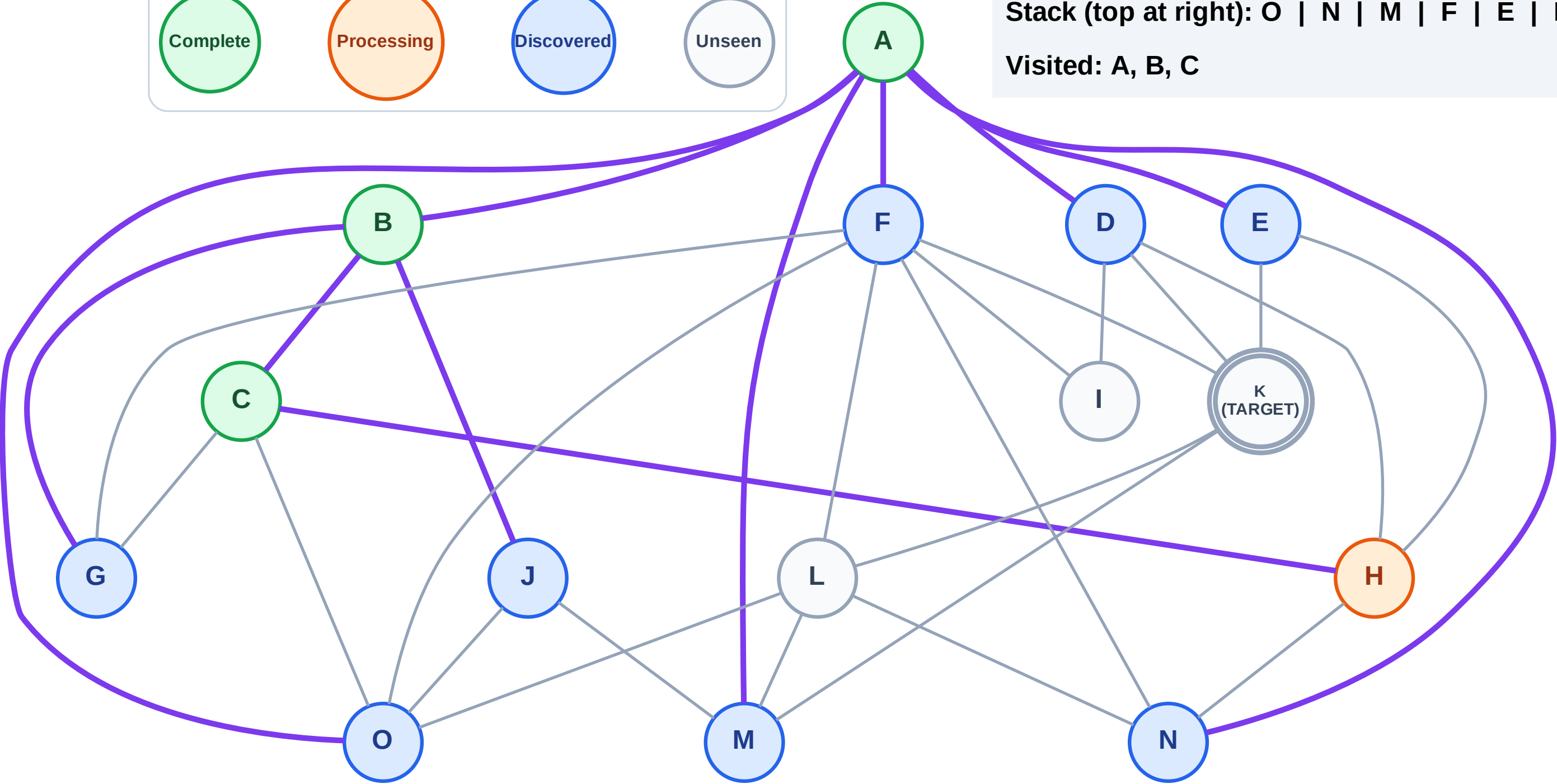
Processing

Discovered

Unseen

Stack (top at right): O | N | M | F | E | D | J | G

Visited: A, B, C



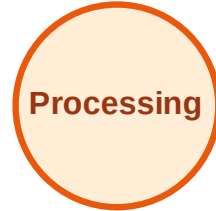
DFS Traversal

Step 9: No new neighbors from H

Legend



Complete



Processing



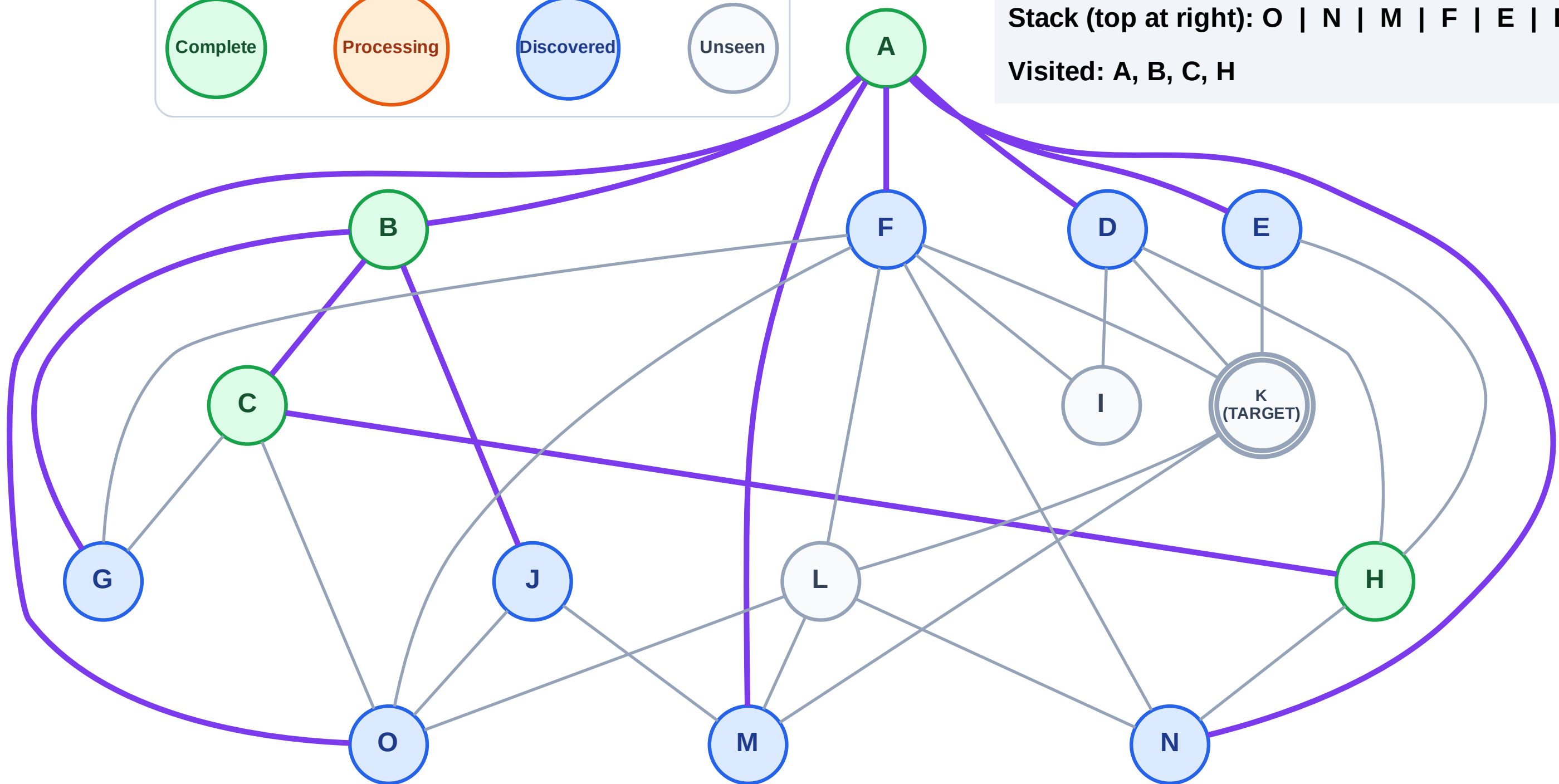
Discovered



Unseen

Stack (top at right): O | N | M | F | E | D | J | G

Visited: A, B, C, H



DFS Traversal

Step 10: Process node G

Legend

Complete

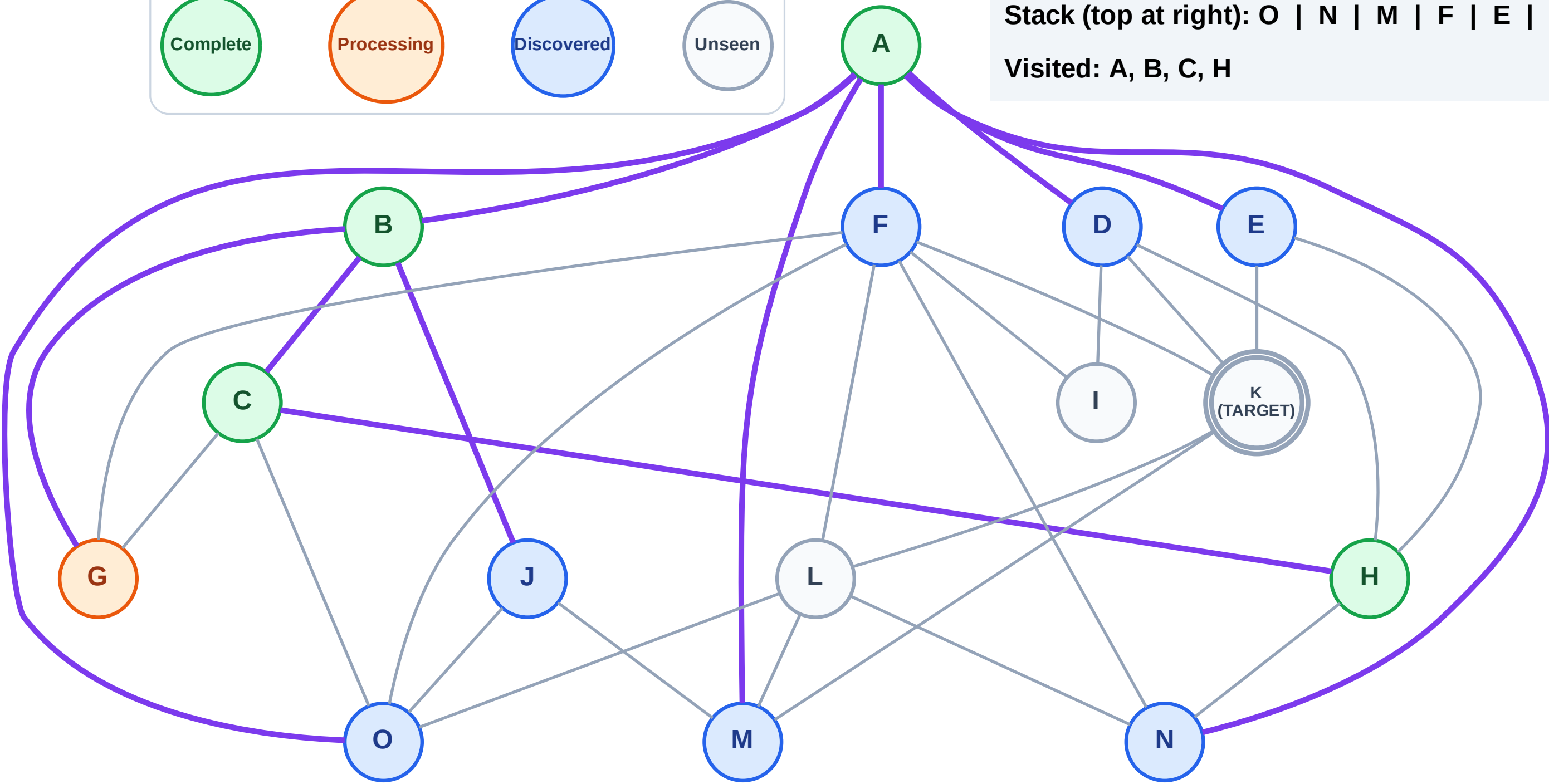
Processing

Discovered

Unseen

Stack (top at right): O | N | M | F | E | D | J

Visited: A, B, C, H



DFS Traversal

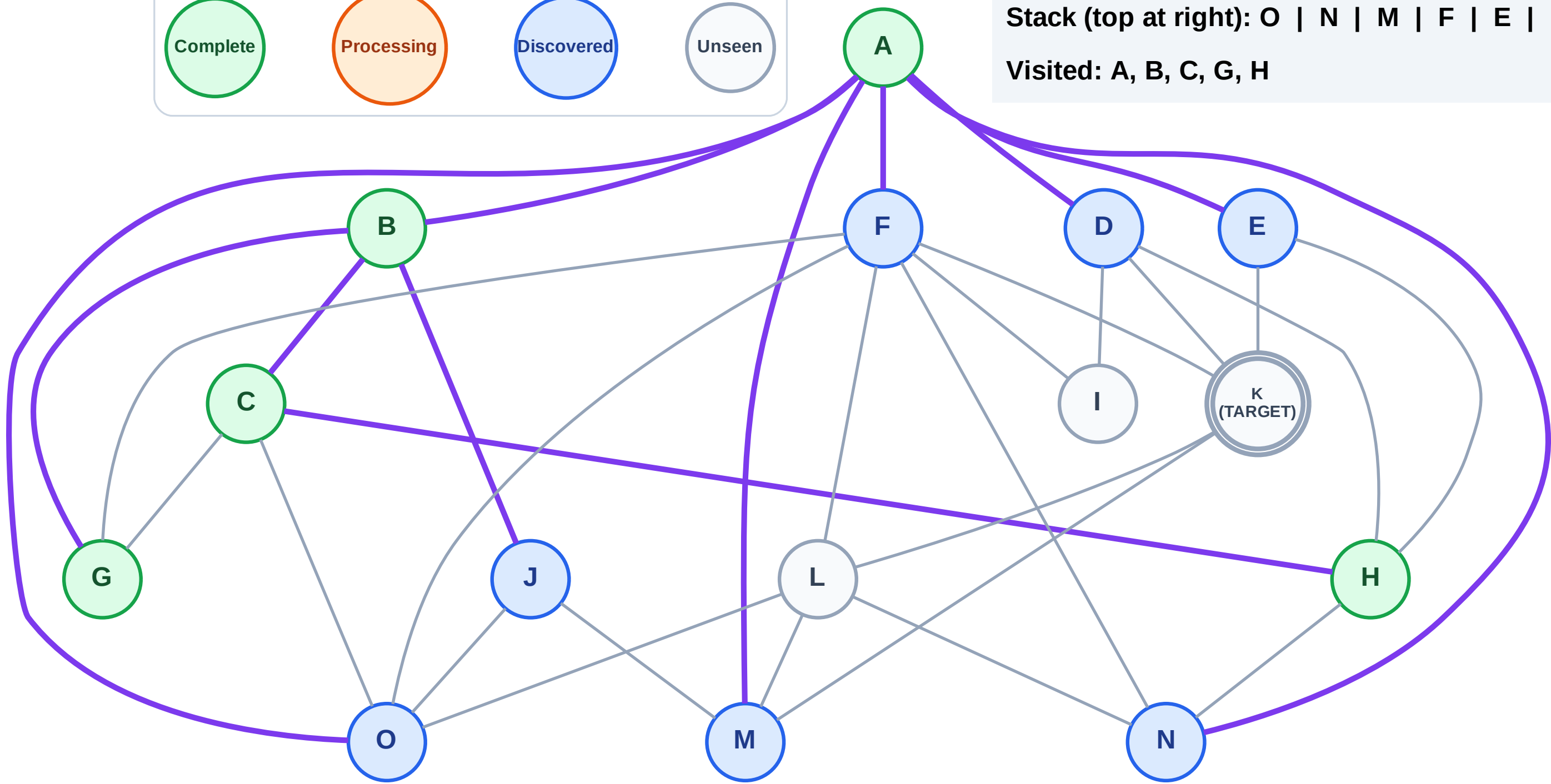
Step 11: No new neighbors from G

Legend



Stack (top at right): O | N | M | F | E | D | J

Visited: A, B, C, G, H



DFS Traversal

Step 12: Process node J

Legend

Complete

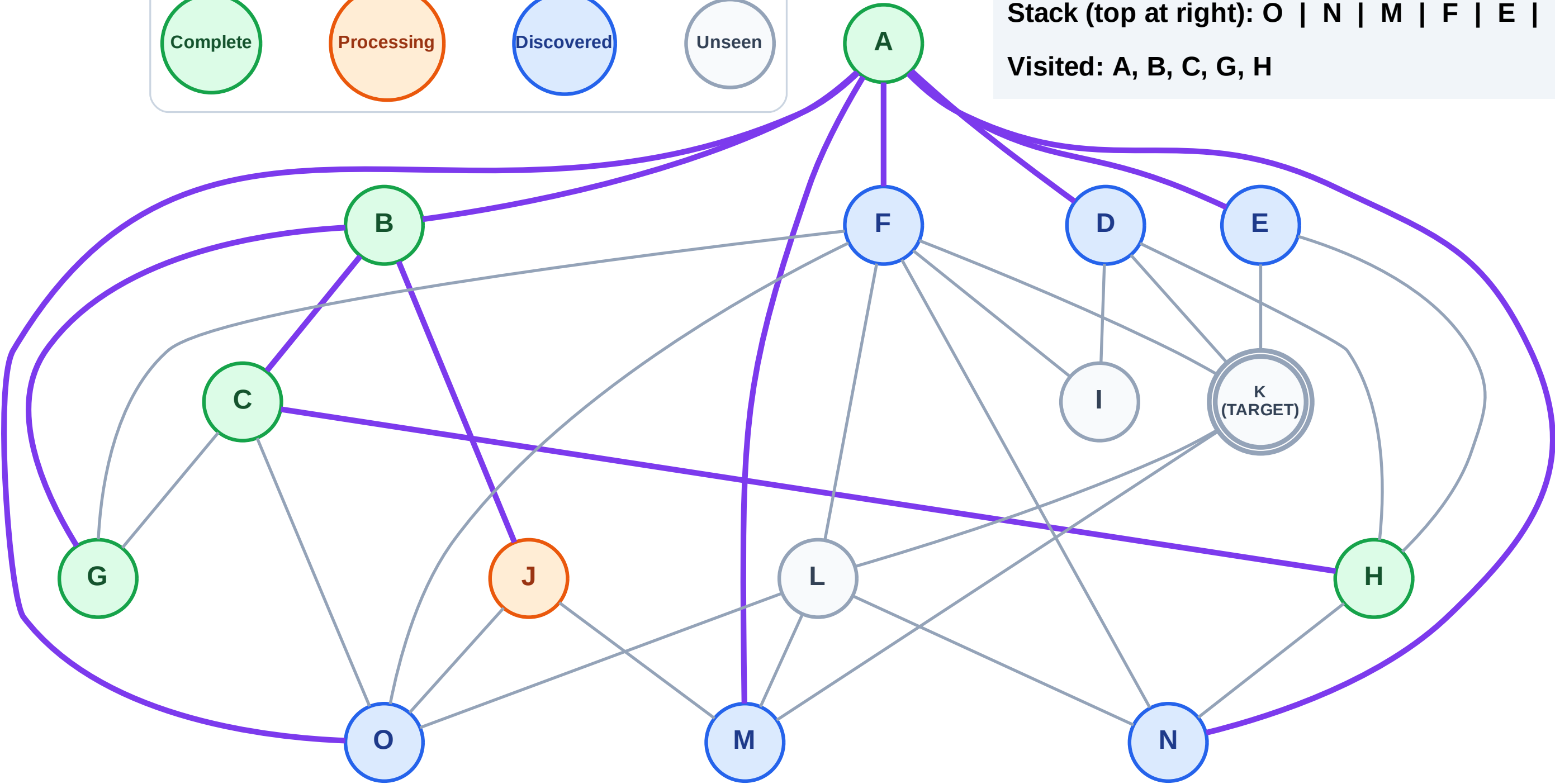
Processing

Discovered

Unseen

Stack (top at right): O | N | M | F | E | D

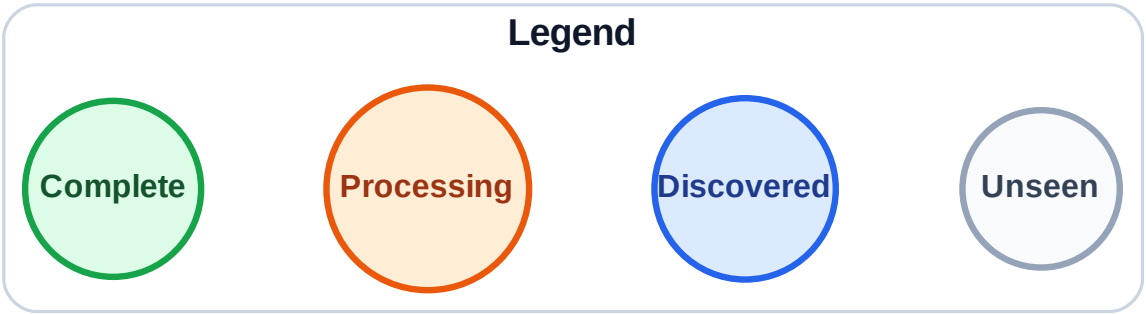
Visited: A, B, C, G, H



DFS Traversal

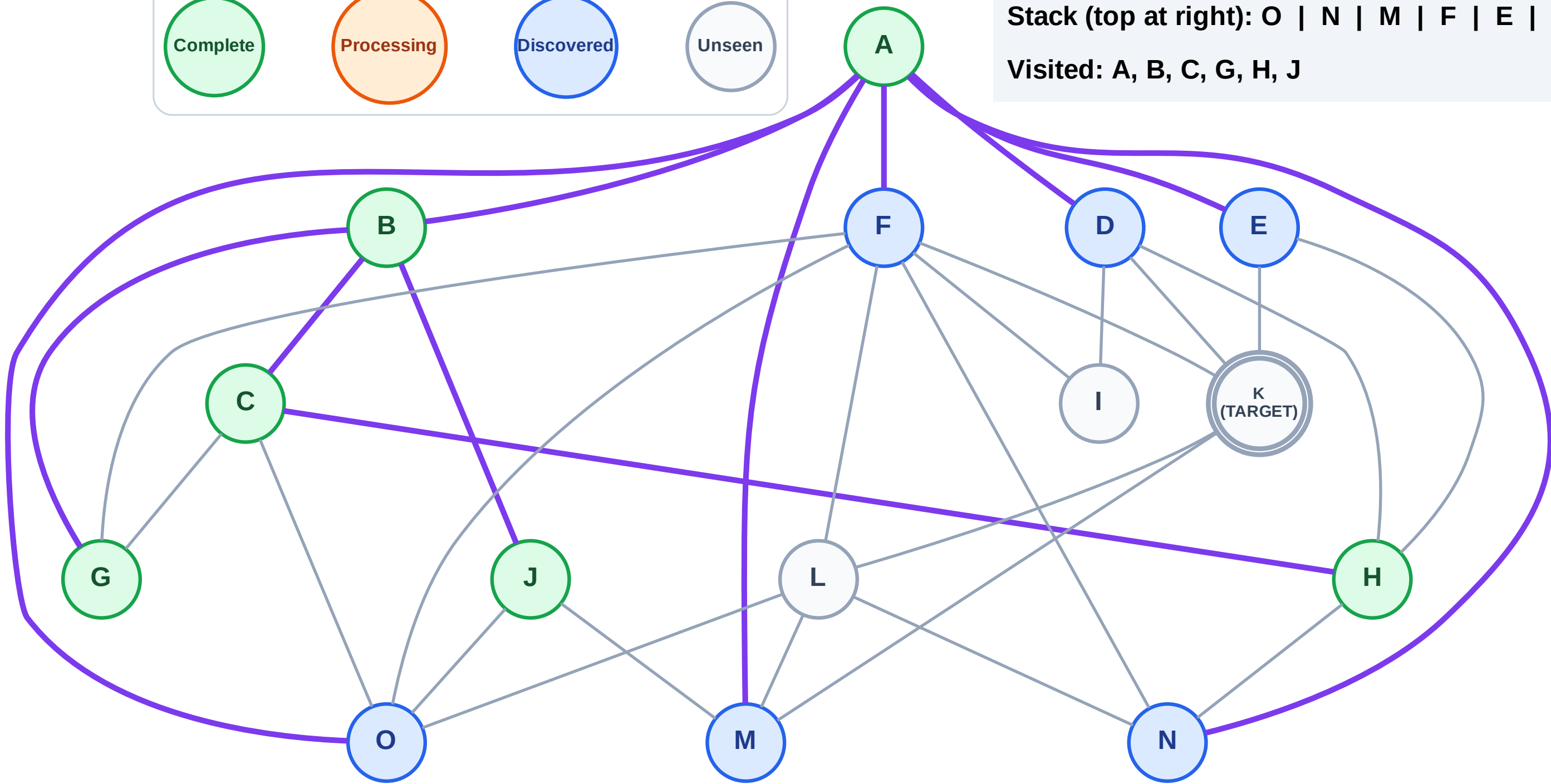
Step 13: No new neighbors from J

Legend



Stack (top at right): O | N | M | F | E | D

Visited: A, B, C, G, H, J



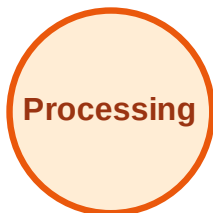
DFS Traversal

Step 14: Process node D

Legend



Complete



Processing



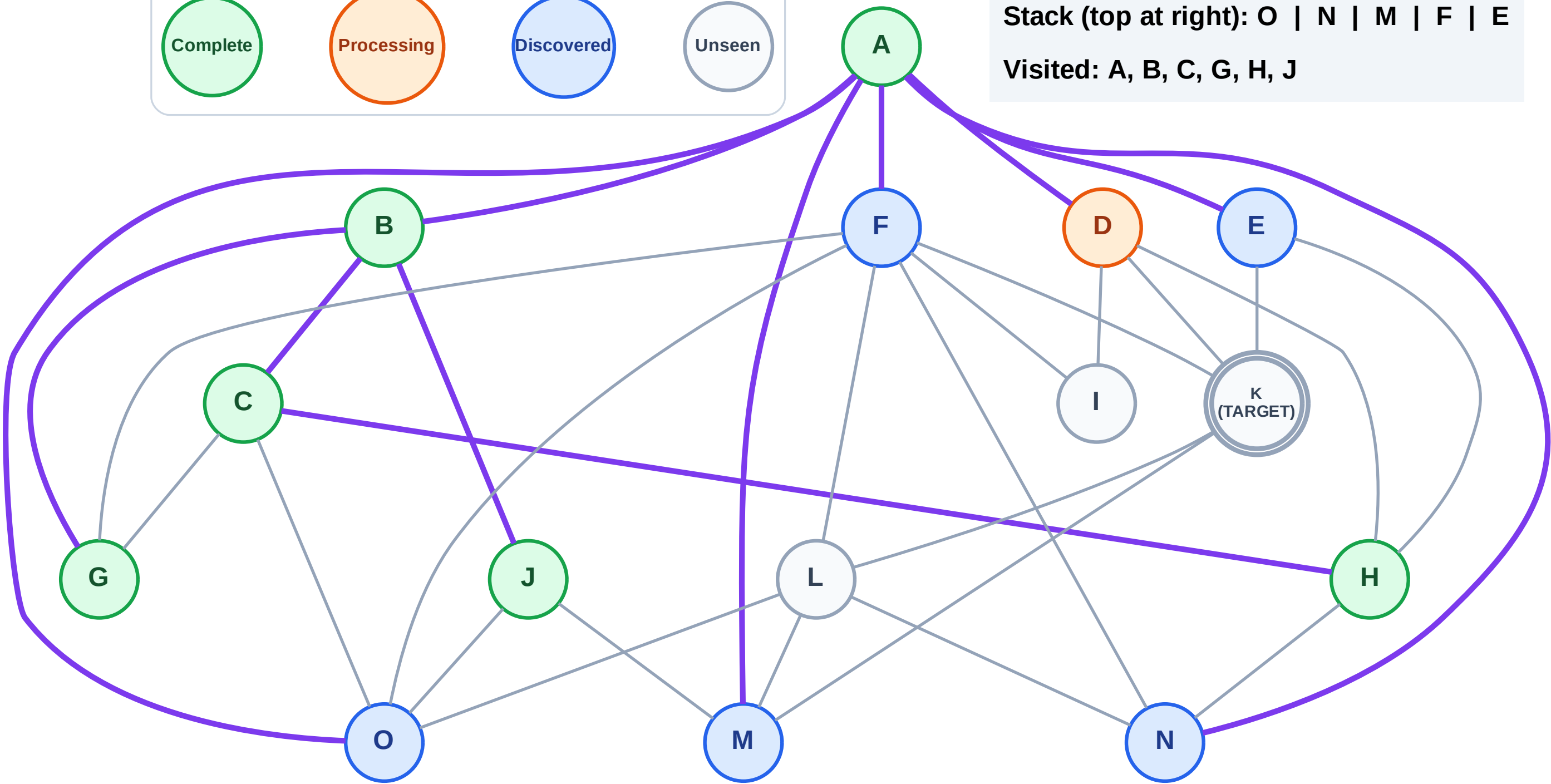
Discovered



Unseen

Stack (top at right): O | N | M | F | E

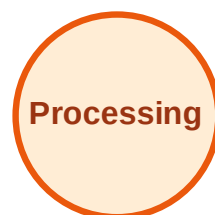
Visited: A, B, C, G, H, J



DFS Traversal

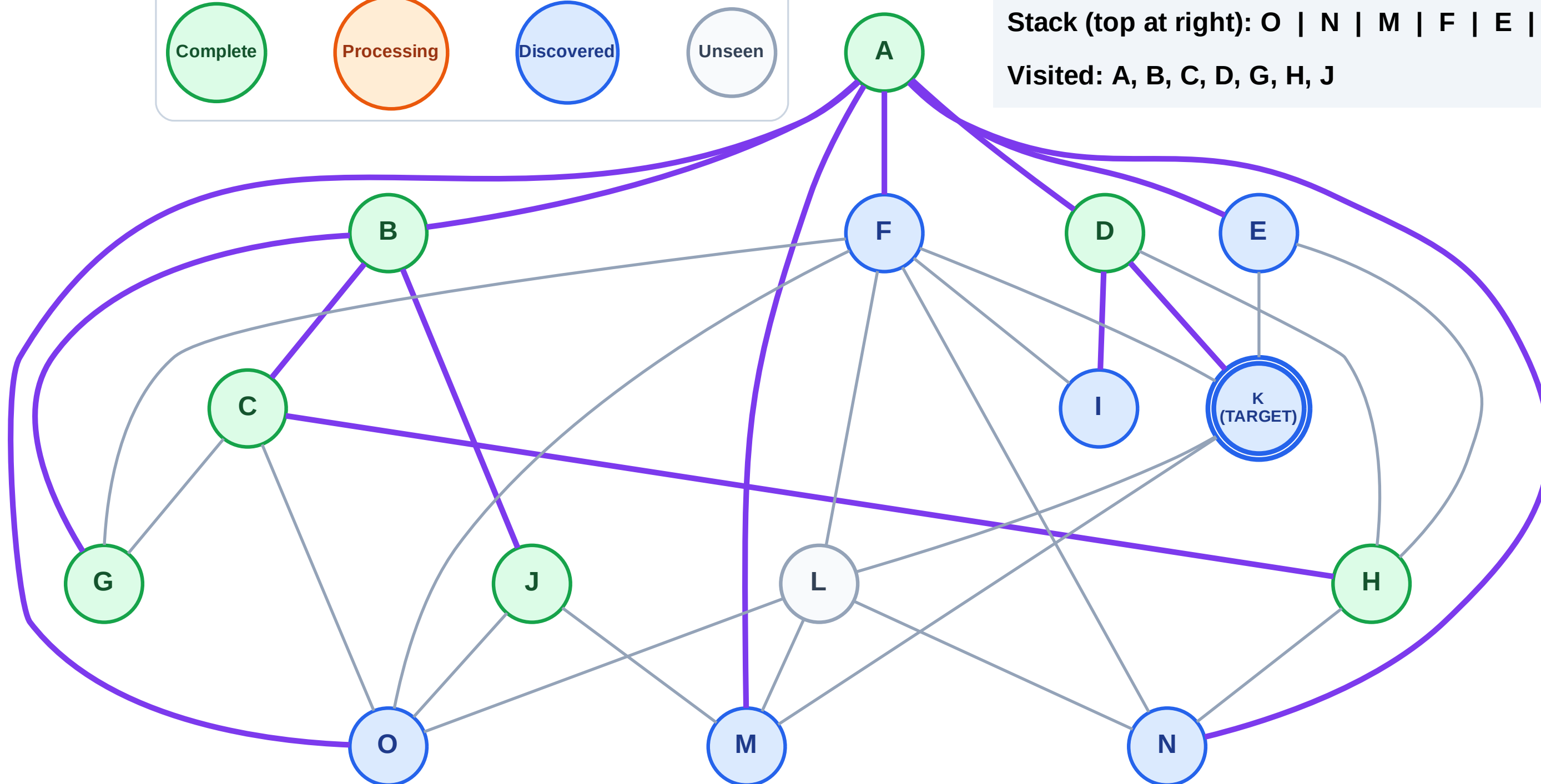
Step 15: Discover K, I from D

Legend



Stack (top at right): O | N | M | F | E | K | I

Visited: A, B, C, D, G, H, J



DFS Traversal

Step 16: Process node I

Legend

Complete

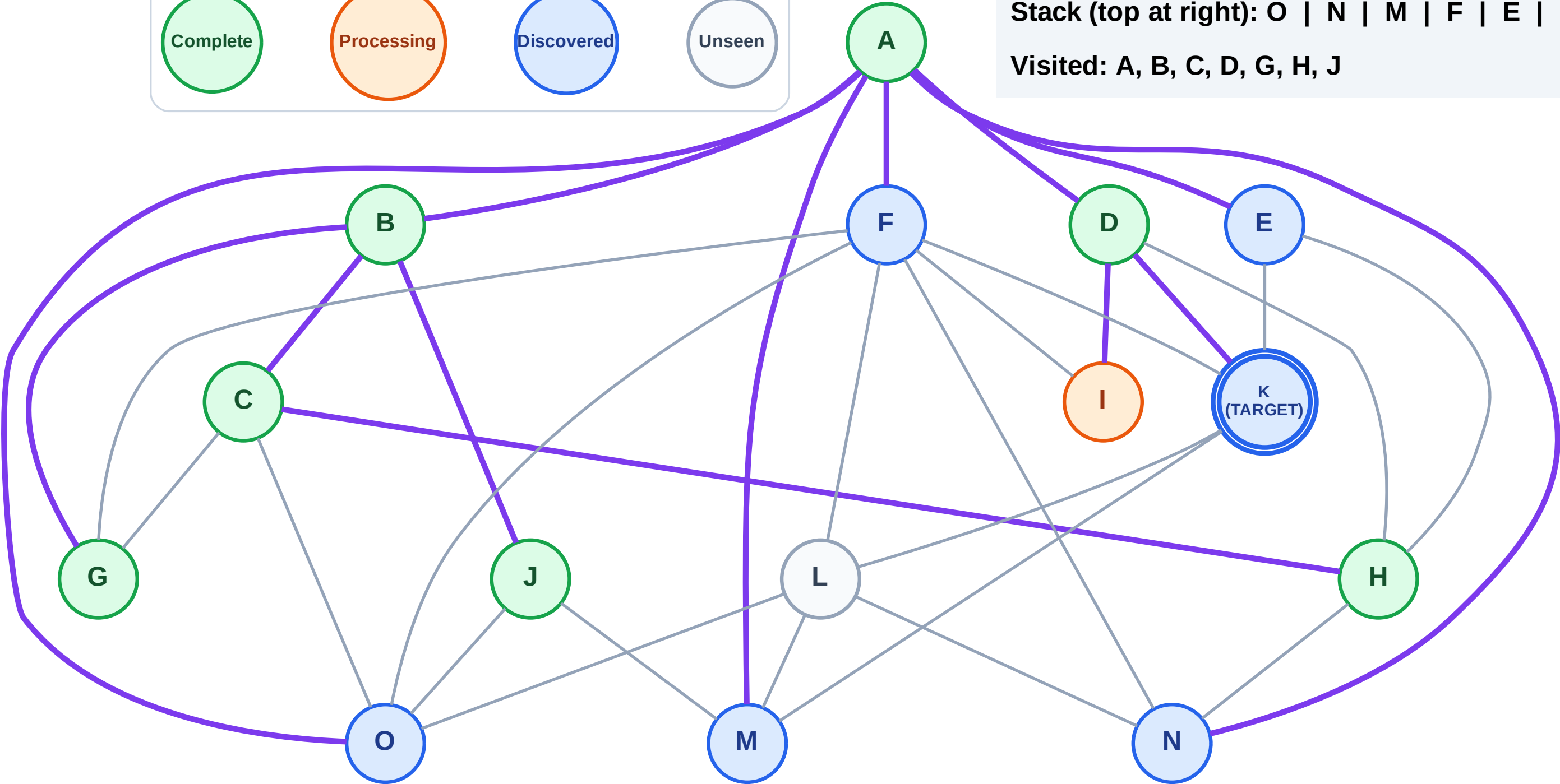
Processing

Discovered

Unseen

Stack (top at right): O | N | M | F | E | K

Visited: A, B, C, D, G, H, J



DFS Traversal

Step 17: No new neighbors from I

Legend



Complete



Processing



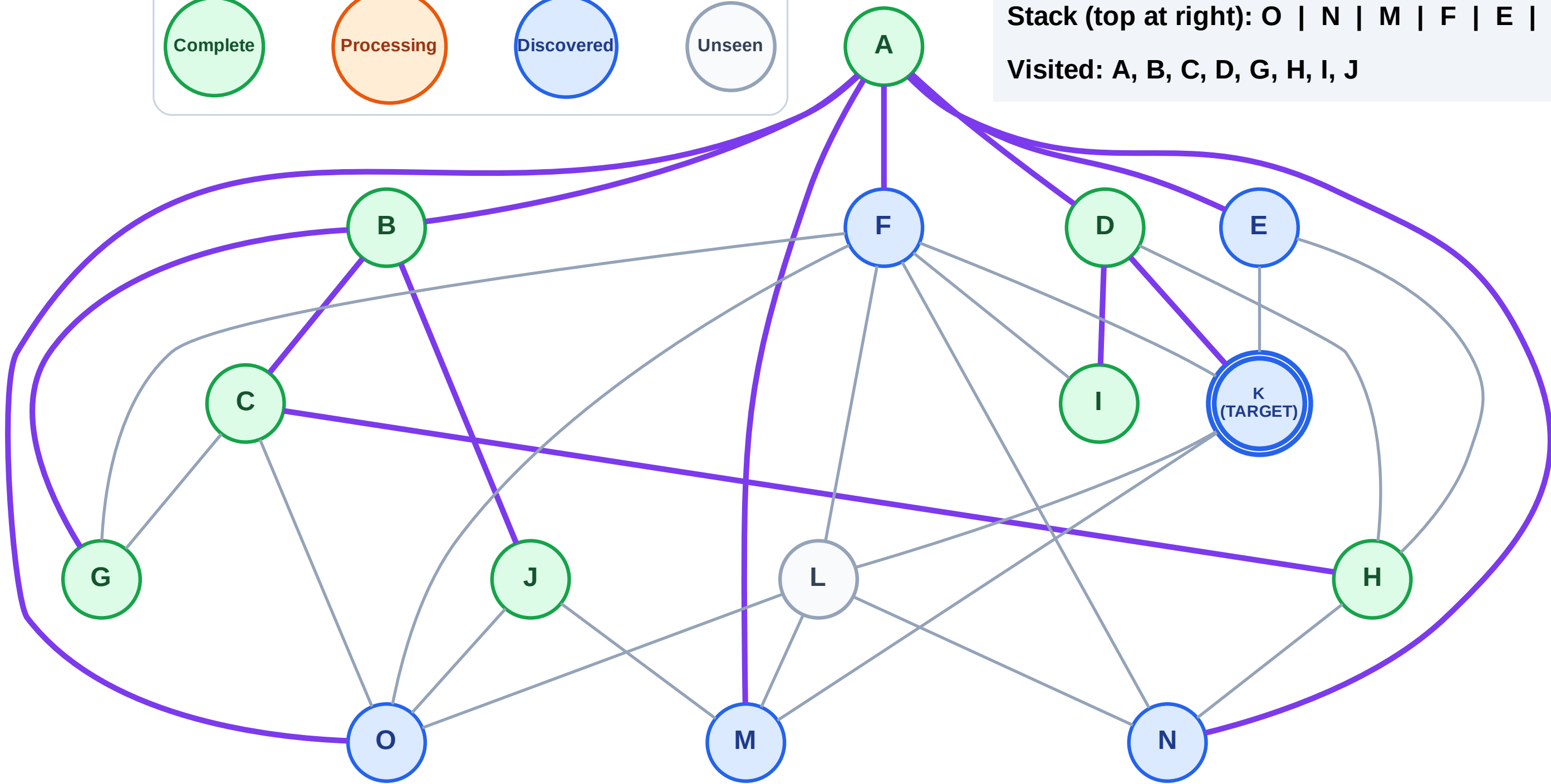
Discovered



Unseen

Stack (top at right): O | N | M | F | E | K

Visited: A, B, C, D, G, H, I, J



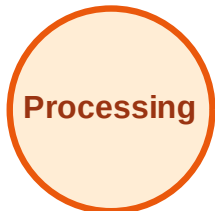
DFS Traversal

Step 18: Process node K

Legend



Complete



Processing



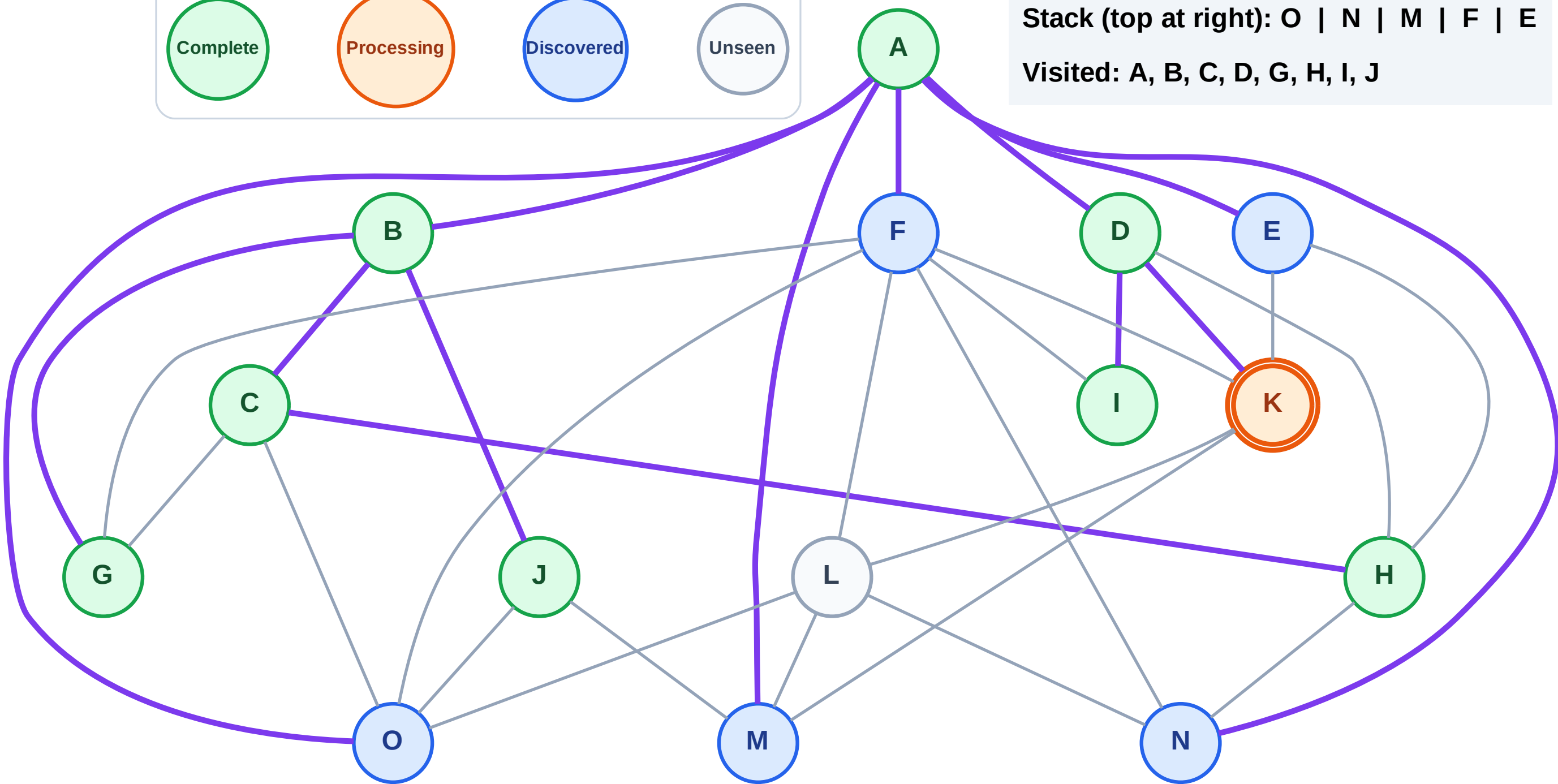
Discovered



Unseen

Stack (top at right): O | N | M | F | E

Visited: A, B, C, D, G, H, I, J



DFS Traversal

Step 19: TARGET NODE K REACHED! Path (2 hops): A → D → K

Legend



Stack (top at right): O | N | M | F | E

Visited: A, B, C, D, G, H, I, J, K

